

ESSENTIAL MATHEMATICS GRADE 10: SIMILARITY AND ENLARGEMENT

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 2.1 (8 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	10
Subject	Essential Mathematics
Strand	Strand 2.0: Measurements and Geometry
Sub-Strand	Sub-Strand 2.1: Similarity and Enlargement
Total Duration	8 lessons × 40 minutes = 320 minutes total
Sub-Strand Content	– Similarity between objects — identifying and sorting similar figures from the environment – Properties of similar figures — corresponding angles and proportional sides – Centre of enlargement — locating and tracing the centre on plain paper – Linear scale factor — measuring corresponding sides and computing the ratio – Drawing the image of an object given the centre of enlargement and linear scale factor – Area scale factor — relating the ratio of image area to object area – Volume scale factor — relating the ratio of volumes of similar solids – Relationship between linear scale factor, area scale factor and volume scale factor – Application of similarity and enlargement in real-life situations – Project: making similar containers (cubes, cuboids, cylinders) of different sizes using locally available materials
Learning Outcomes	By the end of the sub-strand, the learner should be able to: - identify properties of similar figures, - determine the centre of enlargement and the linear scale factor of similar figures, - draw the image of an object given the centre of enlargement and the linear scale factor, - determine the area scale factor of similar figures, - determine the volume scale factor of similar objects, - relate linear scale factor to area and volume scale factor of similar objects, - apply similarity and enlargement to real-life situations, - appreciate the use of similarity and enlargement in real-life situations.
Core Competencies	– Digital Literacy: The learner develops the skills of connecting and interacting with digital technology when using digital resources (e.g., the Kenya Education Cloud or learner digital devices) to manipulate objects or figures — zooming or shrinking sizes — and to learn more on the application of similarity and enlargement in real life. – Creativity and Imagination: The learner acquires networking and connection skills when drawing images of given objects under a particular enlargement scale factor and when designing and constructing similar containers of different sizes during the project.
Core Values	– Responsibility: The learner takes care of objects and containers collected from the immediate environment (e.g., packaging

	tins, juice cartons, sufuria lids) and of instruments used during construction and measurement tasks.
Science & Engineering Practices	– Asking Questions and Defining Problems: Learners observe similar objects from the Kenyan environment (e.g., different-sized Kimbo tins, uji cups, water jerricans) and pose questions about what makes two objects truly similar and how one can be obtained from the other. – Planning and Carrying Out Investigations: Learners systematically measure corresponding sides of similar figures, record data in tables, and compute linear, area, and volume scale factors to test their predictions. – Using Mathematics and Computational Thinking: Learners apply ratios, proportions, and algebraic relationships to determine scale factors and to verify the relationship k^2 (area) and k^3 (volume) from the linear scale factor k. – Constructing Explanations: Learners use measurements and computed scale factors to explain why two containers hold volumes in the ratio k^3 and to justify their model-building decisions during the project. – Obtaining, Evaluating, and Communicating Information: Learners use digital devices and print resources to research real-life applications of similarity and enlargement (e.g., architectural scale drawings, mapmaking, photography) and present findings to peers.
Pertinent & Contemporary Issues (PCIs)	– Environment and Technology: The learner acquires the skill of sorting and responsibly managing the environment after collecting and sorting objects and containers from the immediate surroundings, and uses digital devices appropriately to explore enlargement.
Career Connections	– Architecture and Building Construction: Scale drawings of buildings, rooms, and urban plans in Nairobi or Mombasa rely directly on similarity and linear scale factors. – Cartography and Geography: Map-making for Kenya's counties and national parks uses enlargement and scale factors to represent large real-world distances on paper. – Photography and Media Technology: Zooming in or out on images and resizing photographs for print or digital media applies area scale factor relationships. – Fashion and Textile Design: Pattern grading — scaling clothing patterns up or down for different body sizes — uses the principles of similarity. – Sports Management and Coaching: Designing scaled models of pitches, courts, and arenas (e.g., Kasarani Stadium layouts) involves enlargement and similarity. – Hospitality Industry: Scaling recipes (e.g., ugali, mandazi, pilau) for different numbers of guests and resizing baking tins relies on volume scale factor reasoning. – Engineering and Manufacturing: Producing components in different sizes (e.g., machine parts, water tanks, grain silos in the Rift Valley) requires precise application of linear, area, and volume scale factors.
Focus for Lessons	This unit develops learners' understanding of geometric similarity and enlargement by anchoring instruction in tangible, Kenya-relevant objects — from differently-sized cooking pots and juice containers to architectural scale drawings. Learners move progressively from identifying properties of similar figures, through computing linear, area, and volume scale factors, to constructing their own similar containers as a project. Throughout, digital tools and hands-on measurement tasks are used to build both conceptual understanding and practical competence in applying similarity and enlargement to real-life situations.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: Why does doubling the size of a container make it hold eight times as much — and how can we use this mathematical pattern to design, build, and predict the properties of any similar shape or object? KICD KEY INQUIRY QUESTION: 1. How is similarity and enlargement applied in real life?
Anchoring Phenomenon	ANCHORING PHENOMENON — "The Shrinking Uji Cup":

	A teacher brings to class three different-sized cylindrical uji (porridge) cups — a small cup used by a toddler, a standard adult mug, and a large catering flask — all made by the same manufacturer and clearly sharing the same shape. When learners measure the height and diameter of each cup they discover that every dimension of the larger cup is exactly twice (or three times) the corresponding dimension of the smaller cup. Yet when they fill the small cup with water and try to fill the medium cup, they need FOUR small cups of water — not two. And to fill the large cup they need NINE small cups — not three. The numbers are puzzling: if the cup is "twice as big," why does it hold four times as much water in cross-section and apparently more in volume? Students are also surprised to find that the label (surface area) of the large cup requires NINE times as much paper as the small cup, not three times. This observable, countable mystery — that scaling lengths by k scales areas by k^2 and volumes by k^3 — anchors the entire unit and drives learners to build a mathematical model that explains and predicts these relationships for any similar figures or solids.
Storyline Thread	Lesson 1 — Noticing Similarity (Predict & Observe): Learners collect objects and containers from their immediate environment (tins, cups, boxes). Working in groups, they sort objects they believe are "similar" and record their sorting rules. The teacher introduces the anchoring phenomenon (the three uji cups) and learners predict: if one cup is twice as tall as another, how many small cups of water will fill the big one? Predictions are recorded on the Driving Question Board (DQB). Learners then measure corresponding sides of sorted similar objects, observe that angles are equal and side ratios are constant, and formalise the properties of similar figures. Lesson 2 — Centre of Enlargement and Linear Scale Factor (Observe & Explain): Learners trace an object and its image on plain paper and physically locate the centre of enlargement by drawing lines through corresponding vertices. They measure corresponding sides and compute the linear scale factor k. Multiple examples (different objects, different positions of the centre) consolidate understanding. Learners update the DQB with what they now know about how enlargement works. Lesson 3 — Drawing Enlargements (Explain & Apply): Given a centre of enlargement and a linear scale factor ($k = 2, 3, \frac{1}{2}$), learners construct the image of a given figure step by step. They verify their constructions by measuring. The teacher connects this to real-life contexts: a photographer in Mombasa enlarging a portrait, or a cartographer scaling a map of Kisumu. Learners discuss: does the shape change? Does the angle change? Does size change? They refine their DQB entries. Lesson 4 — Area Scale Factor (Observe & Explain): Learners use cut-out squares and rectangles (similar figures of known dimensions) to compute the area of an object and its image. They tabulate ratios of areas for $k = 2, 3, 4$ and discover the pattern: area scale factor = k^2. The teacher returns to the anchoring phenomenon — the uji cups — and learners explain why four small cups of water fill the cross-section of the medium cup. Model revision: learners add the k^2 relationship to their cumulative class diagram on the DQB. Lesson 5 — Volume Scale Factor (Observe & Explain): Learners use model cubes and cuboids of different sizes (or containers such as small and large Kasuku tins) to measure dimensions, compute volumes, and tabulate the ratio of volumes for $k = 2, 3$. They discover the pattern: volume scale factor = k^3. They explain the anchoring phenomenon fully — why the large uji cup needs nine small cups of water. Learners also connect to the sufuria supporting phenomenon. DQB is updated with k^3. Lesson 6 — Relating Linear, Area and Volume Scale Factors (Explain & Model Building): Learners consolidate all three relationships (k, k^2, k^3) using a unified table. They solve mixed problems: given that two similar water tanks in the Rift Valley have a linear scale factor of 3, find how many times more paint is needed for the larger tank ($k^2 = 9$) and how many times more water it holds ($k^3 = 27$). Digital devices are used to explore and verify. Learners build a complete annotated model (diagram + formula summary) in their notebooks and on the class DQB.

	Lesson 7 — Real-Life Applications and Problem Solving (Apply & Communicate): Learners apply similarity and enlargement to multi-step real-life problems: scale drawings of school buildings, resizing recipes (e.g., scaling a mandazi recipe for a school party — doubling the tin size means eight times the dough), and map reading. Learners use digital or print resources to find additional examples and present one finding to the class. The driving question is revisited and learners articulate a full answer supported by evidence. Lesson 8 — Project Presentations and Assessment (Model Building & Evaluation): Learners present their project: similar containers (cubes, cuboids, cylinders) constructed from locally available materials in at least two different sizes. They demonstrate measurement of dimensions, calculation of linear, area, and volume scale factors, and document challenges. Peers and the teacher use a rating scale for assessment. The class DQB is reviewed and completed — all sub-questions raised in Lesson 1 are answered. A written class activity consolidates individual assessment of all learning outcomes.
Supporting Phenomena	– Differently-sized sufuria (cooking pots) used in a Kenyan kitchen: learners observe that a sufuria with double the diameter holds roughly eight times as much ugali, prompting the need to relate linear scale factor to volume scale factor. – Aerial vs. ground-level photographs of Uhuru Park, Nairobi: a photograph taken from a drone at twice the altitude produces an image in which the park appears one-quarter the size — illustrating the area scale factor k^2 in a real mapping context. – Architectural scale drawings of a school classroom: a builder's blueprint drawn at a scale of 1:50 shows how every length on paper relates to the real dimension, and learners calculate actual room areas and volumes from the drawing, reinforcing all three scale factors. – Packaging of Cowboy or Kasuku cooking fat: the small, medium, and large tins are geometrically similar; learners compare their masses (proportional to volume) and label areas to verify k^2, k^3 relationships using real consumer products found in Kenyan shops.

ATTRIBUTION AND LICENSING

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Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

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LESSON 1 (40 minutes): Noticing Similarity — What Makes Two Shapes the "Same but Different"?

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners encounter the anchoring phenomenon — the three differently-sized uji cups — make and record predictions about how much water each holds, then sort and measure everyday objects to formalise the two defining properties of similar figures: equal corresponding angles and constant ratios of corresponding sides.
Knowledge	– Define similar figures as shapes that have equal corresponding angles and proportional corresponding sides. – State that the constant ratio of corresponding sides is called the linear scale factor (k). – Identify pairs of similar figures from a collection of everyday objects and containers.
Skills	– Measure corresponding lengths of objects using a ruler and record data in a table. – Calculate the ratio of corresponding sides to determine whether two figures are similar. – Sort a collection of objects into 'similar' and 'not similar' groups and justify the sorting rule.
Attitudes	– Show curiosity and open-mindedness when predictions about water volume do not match observations. – Appreciate that mathematical patterns — such as similarity — are embedded in everyday Kenyan objects and manufactured goods.
Key Inquiry Question	How is similarity and enlargement applied in real life?
Purpose in Storyline	This opening lesson launches the unit storyline by placing the anchoring phenomenon — three same-shape uji cups of different sizes — in front of learners. It surfaces prior knowledge through prediction, establishes the two formal properties of similar figures through hands-on measurement of familiar objects, and opens the Driving Question Board with the central mystery: if a cup is twice as tall, why does it hold four times as much water?
Safety Notes	Handle glass or ceramic cups carefully to avoid breakage. If plastic cups are used, ensure they are clean. Water used for the volume demonstration should be disposed of in the sink or outside — not left as a slip hazard on the floor.

B. LESSON OVERVIEW

This is the first lesson of the Similarity and Enlargement unit and its primary purpose is to launch the anchoring phenomenon and open the mathematical inquiry that will run through all subsequent lessons. The teacher introduces three cylindrical uji (porridge) cups — a small toddler cup, a standard adult mug, and a large catering flask — all clearly manufactured to the same shape by the same maker, so that every dimension of the medium cup is exactly twice that of the small cup and every dimension of the large cup is exactly three times that of the small cup. Before any measurements are taken, learners predict in writing how many small cups of water will be needed to fill the medium and large cups. Most learners predict two and three respectively, mirroring the linear scale. The dramatic moment comes when the teacher actually fills the cups with water and learners count four small cups to fill the medium and nine to fill the large — numbers that contradict every intuitive prediction and immediately generate genuine mathematical puzzlement.

The lesson then grounds this puzzlement in the formal language of similarity. Working in groups of four, learners collect everyday objects from around the classroom and their bags — exercise book covers, tins of Kimbo or Blue Band, matchboxes, mobile phone boxes — and sort them into groups they believe are "similar." Each group records the rule they used to decide similarity. The teacher guides groups to test their rules by measuring corresponding sides with a ruler and comparing corresponding angles with a protractor or the corner of a sheet of paper. Through this structured measurement activity learners discover for themselves that similar figures share two measurable properties: corresponding angles are equal and the ratios of corresponding sides are constant. These two properties become the formal definition that the

class agrees on and records.

By the end of the lesson learners have encountered the phenomenon, articulated an initial (and incorrect) prediction, begun to see why the prediction fails, and established the foundational vocabulary — similar figures, corresponding sides, corresponding angles, linear scale factor — that every later lesson will build upon. The Driving Question Board is opened with learner-generated questions, and learners produce their first annotated sketch model showing two similar rectangles with labelled corresponding sides and the ratio calculation. This model will be revised in every subsequent lesson as understanding deepens.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Corresponding points and sides of scaled shapes Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Corresponding sides and points Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Each learner receives a sticky note. The teacher displays the three uji cups at the front of the room and asks learners to study them silently for 30 seconds, then write TWO predictions: (1) How many small cups of water will fill the medium cup? (2) How many small cups of water will fill the large cup? Learners write their predictions individually, fold the sticky note, and hold it until instructed to share. They then share predictions with a shoulder partner before the teacher collects a class sample by cold-calling.	Place the three cups on a tray at the front where every learner can see them clearly. Say: 'Look carefully at these three uji cups. They were made by the same factory — same shape, just different sizes. The medium cup is exactly twice as tall and twice as wide as the small cup. The large cup is exactly three times as tall and three times as wide.' Pause 10 seconds. Then say: 'On your sticky note, write your name and answer these two questions: One — how many small cups of water will completely fill the medium cup? Two — how many small cups of water will completely fill the large cup? Do not discuss yet. You have 60 seconds.' [WAIT 60 seconds in silence.] Then say: 'Now turn to your shoulder partner and compare your answers. You have 30 seconds.' [WAIT 30 seconds.] Cold-call 6 learners (aim for a spread of predictions): 'Amina, what did you write for the medium cup?' — 'Brian, do you agree or disagree with Amina, and why?' — repeat for 4 more learners. Record all unique prediction values on the board under the heading PREDICTIONS — DO NOT ERASE. Do not confirm or deny any prediction. Say: 'We are going	Predict-before-you-observe (Anchored Prediction): Requiring a written, committed prediction before evidence is seen forces learners to make their current mental model explicit. When evidence later contradicts the prediction, cognitive conflict is maximised because every learner has a personal stake in the discrepancy. Recording all predictions publicly on the board validates all thinking and communicates that the goal is sense-making, not being correct.	Teacher scans sticky notes as learners write. Look for: (a) whether learners predict 2 and 3 (linear scaling — the expected misconception), (b) whether any learner predicts 4 and 9 (correct — note for later discussion), (c) whether learners give non-integer answers (shows proportional reasoning). The distribution of predictions reveals the class's prior understanding of scaling and flags the misconception to address explicitly in the Observe phase.

		to test these predictions right now. Keep your sticky note — you will need it later.'		
Observe Phase  VIDEO: Corresponding points and sides of scaled shapes Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Corresponding Parts of Similar Figures (Flexbook) Source: CK-12  Search ARES for similar readings	The teacher fills the small cup with water and pours it into the medium cup, counting aloud with the class. Learners observe and tally on a mini-observation table in their notebooks: Cup pair Predicted fills Actual fills Surprised? (Y/N). The teacher repeats for the large cup. Learners then move into groups of four and collect 4–6 everyday objects from around the room and their bags. Using rulers, they measure the length and width of corresponding faces of pairs of objects they believe are similar and record results in a measurement table: Object A length Object A width Object B length Object B width Ratio L(B)/L(A) Ratio W(B)/W(A). Groups also compare corresponding angles using the corner of a sheet of paper as a right-angle checker.	Hold up the small cup. Say: 'I am going to fill this small cup with water and pour it into the medium cup. Everyone count with me — out loud — each time I pour.' Fill and pour. Count with the class. After the fourth pour fills the medium cup say: 'Four. Not two.' [Pause 15 seconds — do not explain, allow silence and reactions.] Then say: 'Now let us do the large cup.' Repeat pouring from the small cup into the large cup, counting to nine. Say: 'Nine. Not three. Your predictions are on the board. Look at them.' [Pause 10 seconds.] Say: 'In your notebook, complete the observation table I have written on the board. Then write one sentence beginning: I am surprised because...' [WAIT 90 seconds.] Cold-call 3 learners to read their sentence. Then say: 'Hold on to that surprise — it is the question that will drive this entire unit.' Transition to group activity: 'Now I want each group to collect objects from around you — tin cans, boxes, books, anything with a clear rectangular or circular face. Pick pairs you think look similar — same shape, different size. Measure them. I will come around.' Circulate to all groups within 8 minutes. At each group ask: 'What ratio did you get for the lengths? What ratio for the widths? Are they the same or different?'	Discrepant Event followed by Structured Data Collection: The live pouring demonstration creates a discrepant event — an observable outcome that violates a strongly held prediction. The follow-up measurement activity shifts learners from passive surprise to active evidence-gathering, channelling the cognitive conflict into a structured search for pattern. Recording ratios in a two-column table (ratio of lengths vs. ratio of widths) makes the constancy of the ratio visible as data, not just assertion.	During group measurement, the teacher checks: (a) Are learners measuring corresponding sides (length-to-length, width-to-width) or mixing dimensions? (b) Are calculated ratios consistent within a pair (both ≈ 2.0 , or both ≈ 1.5)? (c) Can a learner in the group articulate verbally why constant ratios matter? A group that has inconsistent ratios has likely picked a non-similar pair — use this as a teaching moment: 'Are you sure these two shapes are the same shape?' Groups that finish early are prompted: 'Now check the angles at each corner — are they equal?'
Explain Phase  VIDEO: Using similarity to estimate ratio	Each group shares one pair of objects they measured and their ratio results. The class collates findings on a shared class table on	Bring the class back together. Say: 'Each group, give me one sentence: what ratio did you find, and do you think your two objects	Class Data Synthesis with Formal Definition Anchoring: Pooling group measurements into a single class table makes the pattern	Exit check: ask learners to quickly sketch two rectangles where one has sides 3 cm $\times$ 2 cm and the other has sides 6 cm $\times$ 4 cm, and

between side lengths <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Corresponding Parts of Similar Figures Practice Source: CK-12  Search ARES for similar readings	the board. Learners identify that for all pairs they agreed were 'similar,' the ratio of corresponding lengths was constant and the corresponding angles were equal. The teacher formalises these two properties and introduces the term linear scale factor. Learners write the formal definition in their notebooks and label a diagram of two similar rectangles with the scale factor. They connect this back to the uji cups: the cups have a linear scale factor of 2 (medium:small), yet the water fills were 4 — a number the teacher asks learners to notice equals 2^2.	are similar?' Cold-call all 6 groups in turn. Record each group's ratio on the board class table. Say: 'Look at this table. What do you notice about the groups whose objects you all agree look similar?' [WAIT 15 seconds.] Invite 3 learners: 'Zawadi, what do you see?' After responses, say: 'The pattern you have found has a name. When two figures are similar, their corresponding angles are equal AND the ratios of their corresponding sides are equal. That constant ratio is called the LINEAR SCALE FACTOR — we write it as k.' Write on the board: Similar figures $\rightarrow$ (1) Corresponding angles equal, (2) Corresponding sides in constant ratio k. Say: 'Now look at the uji cups. The medium cup has a linear scale factor of $k = 2$ compared to the small cup. We filled it with 4 small cups of water. Someone tell me: what is 2 squared?' [WAIT 10 seconds.] Cold-call 2 learners. Say: 'And the large cup: $k = 3$, fills = 9. What is 3 squared?' [WAIT 10 seconds.] Say: 'Write this in your notebook and circle it: $k = 2 \rightarrow$ fills = $4 = 2^2$. $k = 3 \rightarrow$ fills = $9 = 3^2$. We are not ready to explain this fully yet — but keep it in mind. That is our puzzle for the unit.'	(constant ratio = similar) emerge from the learners' own data rather than being imposed by the teacher. Formal vocabulary is introduced only after the pattern is visible in the data, so the definition is a name for something learners have already noticed. Connecting the formal definition immediately back to the phenomenon ($k = 2$, fills = $4 = k^2$) plants the seed of the area-scale-factor relationship without requiring learners to explain it yet.	write whether they are similar and what the linear scale factor is. Circulate and check: (a) Does the learner correctly identify $k = 2$? (b) Does the learner correctly identify that angles are equal (all 90°)? (c) Does the learner check both ratios ($6/3 = 2$ AND $4/2 = 2$)? This takes 2 minutes and reveals whether the formal definition has been internalised before the DQB phase.
Driving Question Board (DQB) Creation  VIDEO: Dilations and properties <i>Related topic: not an exact match for this lesson.</i> Source: Khan	Each learner writes one question on a sticky note prompted by what they observed or cannot yet explain — especially about the uji cup mystery. Sticky notes are posted on a dedicated section of the wall or board labelled DRIVING QUESTION BOARD. The teacher reads selected questions aloud and the class organises them into two columns:	Distribute a new sticky note to each learner. Say: 'Think about everything you saw and measured today. Write one question — something you observed that you cannot yet explain, or something you want to understand by the end of this unit. Start your question with WHY, HOW, or WHAT IF.' [WAIT 2 minutes in silence.] Then say: 'Come and post your question	Driving Question Board (DQB) — Public Knowledge Tracker: The DQB makes the class's collective knowledge state visible and persistent. It externalises metacognition: learners can see what the class knows and what remains mysterious. Returning to the DQB at the start and end of every lesson creates a storyline thread — learners experience their	Teacher reads all sticky notes before the next lesson and categorises them: (a) questions about area scaling (k^2) — will be addressed in Lesson 3; (b) questions about volume scaling (k^3) — will be addressed in Lesson 5; (c) questions about what 'corresponding' means — address in Lesson 2 warm-up; (d) questions that reveal a continuing

Academy (English - US curriculum) Search ARES for similar videos HTML: Corresponding Sides of Similar Triangles (PLIX) Source: CK-12 Search ARES for similar readings	Things we have noticed and Things we still cannot explain. The central driving question — 'Why does doubling the size of a container make it hold eight times as much?' — is written in large text at the top of the DQB by the teacher if not already suggested by a learner.	on the board.' Once all notes are posted, read 5–6 aloud, grouping similar questions by placing them close together. Say: 'I notice many of you are asking about the water cups — why four, why nine. That is our central mystery.' Write on the board in large letters: DRIVING QUESTION — Why does doubling the size of a container make it hold eight times as much — and how can we use this pattern to design and predict the properties of any similar shape? Say: 'Every lesson this unit, we will come back to this board. As we answer questions, we will move them to the EXPLAINED side. Questions that are still open stay here. By the end of the unit, this board should be mostly on the EXPLAINED side.' Point to the two columns you have labelled: STILL WONDERING WE CAN NOW EXPLAIN.	own growing understanding as a narrative, not as isolated facts. Anchoring the DQB to the phenomenon ensures that abstract mathematical learning is always connected to a concrete, memorable context.	misconception (e.g. 'Is $k = 2$ because the cup is twice as tall?') — flag for explicit address in Lesson 2. This informs the teacher's planning for subsequent lessons.
Model Building Phase VIDEO: Side lengths after dilation <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos HTML: Unknown Measures of Similar Figures (Flexbook) Source: CK-12 Search ARES for similar readings	In their notebooks, learners draw and label their first unit model: two similar rectangles (one with sides $3\text{ cm} \times 2\text{ cm}$, one with sides $6\text{ cm} \times 4\text{ cm}$), annotated with arrows showing corresponding sides, the ratio calculation written beside each pair of corresponding sides, and $k = 2$ written prominently. Below the rectangles, learners sketch the three uji cups (labelled small, medium, large) and write the linear scale factors $k = 2$ and $k = 3$ beside the medium and large cups. They leave a blank box labelled '? fills' under each cup — to be completed in a future lesson when they understand why fills = k^2.	Say: 'In your notebook, on a fresh page, draw your first model for this unit. It has two parts. Part one: draw two similar rectangles — the small one is 3 cm by 2 cm. Use your ruler. The large one has $k = 2$. Draw it beside the small one. Label the sides. Write the ratio calculation. Part two: below the rectangles, sketch the three uji cups — label them small, medium, large. Write $k = 2$ beside the medium cup and $k = 3$ beside the large cup. Leave a blank box under each cup and write inside it: ? — we will fill this in as we learn more.' Circulate for 4 minutes. Check that learners: (a) have drawn the large rectangle with sides $6\text{ cm} \times 4\text{ cm}$ (not $5\text{ cm} \times 4\text{ cm}$ or other), (b) have written ratio calculations, (c) have k correctly labelled. Say: 'This model	Cumulative Annotated Model (CAM): Requiring learners to build a single evolving model — rather than isolated diagrams per lesson — creates a physical representation of the unit storyline in every learner's notebook. The deliberate blank boxes ('? fills') embed the unresolved mystery directly into the model, so that the mathematical tension is visible and persistent. Each time a blank is filled in a future lesson, learners experience a concrete sense of progress and explanation.	Teacher does a rapid gallery walk during the 4-minute drawing period. Look for: (a) correct second rectangle dimensions (6×4); (b) ratio written as a fraction or decimal beside each pair of sides; (c) $k = 2$ and $k = 3$ labels on the cup sketches. Learners who have the wrong second rectangle dimensions (common error: adding 2 to each side instead of multiplying) are asked: 'You found $k = 2$ from the data. How did you get $k = 2$ from 3 cm to your new length?' This probes multiplicative vs. additive thinking — the most critical prerequisite misconception for the unit.

		is the beginning. Every lesson we will add to it. By the end of the unit it will explain everything on the Driving Question Board.' Cold-call one learner to hold up their model and describe it to the class in one sentence.		
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D. TEACHER REFLECTION

1. When learners saw that the medium cup needed 4 small cups of water (not 2), how did they react — surprise, disengagement, or immediate curiosity? Did the discrepant event generate the cognitive conflict you hoped for, and if not, how could you strengthen the setup (e.g. asking learners to bet on their prediction before pouring)?
2. During the group measurement activity, did most groups correctly identify similar pairs and find consistent ratios for both length and width? Which groups struggled, and was the difficulty conceptual (not understanding what 'corresponding' means) or procedural (inaccurate measuring technique)?
3. When you cold-called learners to share their DQB questions, did the questions reveal understanding of the area/volume scaling mystery, or were most questions about the definition of similarity itself? What does this tell you about the level at which to pitch Lesson 2?
4. In the Model Building phase, how many learners made the additive error (adding k to each dimension instead of multiplying)? This misconception — that scaling is additive — is the root cause of all later errors in this unit. Plan a specific 5-minute warm-up for Lesson 2 to address it if more than a quarter of the class made this error.
5. Did the three uji cups feel authentic and culturally relevant to learners, or did the context feel contrived? Could you involve learners in bringing similar-shaped containers from home for Lesson 2 to increase ownership and relevance of the phenomenon?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Three uji cups of different sizes (small, medium, large) share the same shape — every dimension of the medium cup is exactly twice that of the small cup ($k = 2$) and every dimension of the large cup is exactly three times that of the small cup ($k = 3$). When filled with water, the medium cup required 4 small cups (not 2) and the large cup required 9 small cups (not 3). In everyday objects measured by groups, similar pairs showed equal corresponding angles and a constant ratio of corresponding sides.
What did I learn?	Two figures are similar if and only if their corresponding angles are equal AND their corresponding sides are in a constant ratio called the linear scale factor (k). Similarity is about shape, not size. The linear scale factor k can be found by dividing the length of any side of the larger figure by the corresponding side of the smaller figure. For the uji cups, $k = 2$ (medium:small) and $k = 3$ (large:small) — but the number of water fills was $k^2 = 4$ and $k^2 = 9$, not k itself. This k^2 pattern is the mystery that the rest of the unit will explain.
How does this explain the phenomenon?	The three uji cups are similar solids — they share the same shape, with all dimensions scaled by the same factor k . This is why they look proportional to the eye. However, the amount of water each holds (related to cross-sectional area and volume) does not scale by k but by a larger power of k . Today's lesson establishes the foundation — the definition of similarity and the meaning of k — which is the first step toward building a mathematical model that predicts the 'surprising' water-fill numbers and explains why

	scaling lengths by k scales areas and volumes by higher powers of k .
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LESSON 2 (80 minutes (2 × 40-minute lessons)): Centre of Enlargement and Linear Scale Factor

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners investigate how enlargement works by physically locating the centre of enlargement and computing the linear scale factor from real traced figures, building the mathematical foundation needed to later explain why the uji cups hold 4× and 9× as much water.
Knowledge	Learners will know: (a) what the centre of enlargement is and how to locate it by drawing lines through corresponding vertices; (b) how to measure corresponding sides of an object and its image; (c) how to calculate the linear scale factor $k = \text{image length} \div \text{object length}$.
Skills	Learners will be able to: (a) trace an object and its image on plain paper and locate the centre of enlargement; (b) measure corresponding sides accurately using a ruler; (c) compute the linear scale factor k from at least three pairs of corresponding sides; (d) draw the image of a simple polygon given a centre of enlargement and a linear scale factor.
Attitudes	Learners will appreciate that enlargement is a precise, rule-governed transformation — not a freehand 'making bigger' — and will develop confidence in using geometric tools to uncover mathematical structure hidden in everyday objects such as the uji cups.
Key Inquiry Question	How is similarity and enlargement applied in real life?
Purpose in Storyline	In Lesson 1 learners noticed the puzzling water-filling results with the uji cups and asked: 'What rule controls how the cups scale?' Lesson 2 answers the first part of that question by revealing the geometric engine behind enlargement — the centre of enlargement and the linear scale factor k . Once learners see that every ray from the centre passes through both the object vertex and the image vertex, and that k tells us exactly how far each point moves, they have the conceptual tool to later explain why area scales as k^2 and volume as k^3 .
Safety Notes	No significant hazards. Remind learners to handle drawing compasses and sharp pencils carefully. Ensure tables are clear of bags to avoid tripping when learners move to compare traced figures.

B. LESSON OVERVIEW

Learners open by revisiting the uji cup mystery from Lesson 1 and predicting what geometric rule might control how the cups were made 'the same shape but different size.' They then carry out a hands-on tracing activity: each pair traces a simple polygon (a cut-out of a Kenyan chai cup cross-section) and its pre-drawn enlarged image, then draws rays through corresponding vertices to pinpoint the centre of enlargement. They measure at least three pairs of corresponding sides and compute k each time, discovering that k is constant for all pairs. A second example with a different object (a Rift Valley tea-leaf outline) placed at a different centre consolidates the concept. Learners close by updating the Driving Question Board and beginning a cumulative anchor model linking k to the cup phenomenon.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Dilations: scale factor Source: Khan	Each learner receives a half-page printed sheet showing two similar triangles — labelled 'Small Uji Cup profile' and 'Large Uji Cup profile' — placed on plain paper with no	Distribute the printed sheet face-down. Say: 'Do NOT flip it over yet.' After 30 seconds of silence, say: 'Flip and read — you have 3 minutes to write your own	Think-Pair-Share with Anchored Prediction: by requiring learners to commit a number and a reason to paper before discussion, the teacher activates prior knowledge	Observable indicator: learner has written a specific numerical prediction (not 'bigger') AND a geometric reason (e.g., 'rays go out from a point,' 'everything

Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Dilations: scale factor Source: Khan Academy (English - US curriculum) Search ARES for similar readings	centre marked. Learners individually write responses to two prompts in their exercise books: (1) 'How do you think the manufacturer moved every point of the small cup outward to create the large cup? Draw an arrow showing your idea.' (2) 'If the large cup's height is 12 cm and the small cup's height is 4 cm, predict what single number describes the relationship between every pair of matching lengths.' Learners share predictions with a shoulder partner for 2 minutes.	prediction before talking to anyone.' [WAIT — 3 minutes individual silent writing.] Then say: 'Now share your prediction with your partner. You have 2 minutes. I want to hear the number you predicted AND the reason.' Circulate and listen; note 2–3 varied predictions on a sticky note for later reference. After partner talk, cold-call 3 learners (choose one who predicted $k = 3$, one who predicted $k = 2$, one who is uncertain): 'Amina, what number did you predict and why?' [WAIT 10 seconds after each nomination before accepting response.] Record all three predictions visibly on the board under the heading 'OUR PREDICTIONS — we will check these today.'	from Lesson 1 (the cup measurements) and creates cognitive dissonance that the Observe phase will resolve. Recording predictions publicly creates accountability and gives the teacher formative data about misconceptions (e.g., learners who think the number is 2 because 'it looks twice as big').	moves the same amount'). Teacher notes which learners write only a number with no geometric reasoning — these learners need targeted questioning during the Observe phase.
Observe Phase VIDEO: Dilations: center Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Dilations: scale factor Source: Khan Academy (English - US curriculum) Search ARES for similar readings	ACTIVITY A — Locating the Centre (15 min): Each pair receives: one A4 plain paper sheet with a pre-drawn pentagon labelled ABCDE (object, side ≈ 3 cm) and its enlarged image A'B'C'D'E' already printed, positioned so the centre of enlargement is on the page but unmarked; one ruler; one sharp pencil. Pairs draw a straight line through A and A', extending it in both directions. They repeat for $B \rightarrow B'$ and $C \rightarrow C'$. They mark the point where all three lines meet and label it O — the centre of enlargement. Learners measure OA and OA', then OB and OB', then OC and OC'. They record the six measurements in a table and compute the ratio OA/OA', OB/OB', OC/OC' for each pair. ACTIVITY B — Computing k from Side Lengths (10 min): Pairs now measure corresponding sides: A'B'/AB, B'C'/BC, C'D'/CD, D'E'/DE,	Before releasing materials say: 'You are now the manufacturer's quality inspector. Your job is to prove — using your ruler and straight lines — exactly where the factory's enlargement machine was centred. This is the secret behind how every uji cup in Kenya is made identical in shape.' Distribute materials. [WAIT — allow 2 minutes for pairs to attempt drawing rays before intervening.] Circulate; for any pair drawing rays that do not converge, crouch to their level and ask: 'Are you drawing through BOTH the object vertex AND the image vertex? Check your pencil is on A first, then slide it to A'.' Do NOT draw the line for them — guide only. After 8 minutes, pause the class: 'Raise your hand if your three lines all meet at one point.' [Count hands aloud.] 'Excellent — Kariuki's group and Fatuma's group, what do you notice about	Hands-On Data Collection with Ratio Table: learners generate their own numerical evidence rather than being told k. The repeated act of measuring five different pairs and obtaining the same ratio each time creates the 'aha' that k is a property of the whole transformation, not just one pair of sides. This directly mirrors how the uji cup manufacturer would verify that every dimension scaled identically.	Observable indicators: (1) Three rays drawn through corresponding vertex pairs that visibly converge at a single point O. (2) A completed two-column ratio table with at least five entries, all values within ± 0.1 of each other. (3) A written sentence correctly identifying k as a constant ratio. Teacher flags any pair whose ratios vary by more than 0.2 for immediate re-measurement support.

	E'A'/EA. They record all five ratios in a second table. They write a sentence: 'The linear scale factor k = ____ because every ratio of image side to object side equals ____.'	where O landed relative to the two pentagons?' [WAIT 10 seconds.] Cold-call 2 more groups. For Activity B, say: 'Now use your ruler — not your eye — to measure EVERY pair of matching sides. Record every measurement. I will be checking that you have at least five ratios in your table.' Circulate and check tables. If a ratio differs from the others by more than 0.1, prompt: 'Re-measure that side — start your ruler exactly at the vertex.'		
Explain Phase  VIDEO: Dilations: scale factor Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Prove triangle similarity <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	CONSOLIDATION EXAMPLE (8 min): Each learner individually draws the image of a simple quadrilateral PQRS (given on a printed grid, side lengths 2 cm each, forming a square) under an enlargement with centre O (marked at a specific grid point) and scale factor $k = 3$. Learners must: (a) draw a ray from O through P and mark P' at distance $OP \times 3$ from O; (b) repeat for Q, R, S; (c) join P'Q'R'S'; (d) verify by measuring one pair of sides. DISCUSSION (12 min): Whole-class structured discussion using the displayed anchor photograph of the three uji cups. Teacher writes on the board: 'Small cup height = 5 cm. Medium cup height = 10 cm. Large cup height = 15 cm. Where is the centre of enlargement for the cup manufacturer?' Learners discuss in groups of 4 for 3 minutes, then share. Teacher records key agreed statements: 'k from small to medium = ____'; 'k from small to large = ____'; 'Every length on the medium cup is ____ times the matching length on the small cup.'	For the individual drawing task, say: 'Close your books. No partner help for 8 minutes. This is your chance to show me YOU can do this.' [WAIT — enforce individual silence.] Circulate and observe; mark a small tick in pencil next to any correctly placed image vertex. After 8 minutes say: 'Swap books with your partner. Check each other's P', Q', R', S' using your ruler. If a point is more than 2 mm off, mark it with a question mark — do not erase your partner's work.' [WAIT 2 minutes.] Open the whole-class discussion by displaying the uji cup photo. Ask: 'Baraka, if you were standing at the centre of enlargement, what would you see?' [WAIT 12 seconds — this is deliberately open.] Then: 'Adhiambo, how does the scale factor k we measured today help explain why the manufacturer can guarantee every cup is the same shape?' [WAIT 10 seconds.] Cold-call one more learner: 'Mwangi, complete this sentence on the board for us: The linear scale factor k tells us that every length on the image is ____.' Correct any statement that says 'k tells us the size' — redirect: 'k tells	Individual Application followed by Peer Verification and Whole-Class Anchoring: the individual drawing task forces each learner to operationalise the procedure (not just observe it). Peer verification builds checking habits. The return to the uji cup anchors the abstract procedure in the phenomenon, ensuring the Explain phase does not float free of the unit's driving question.	Observable indicators: (1) Correctly constructed image P'Q'R'S' with all four vertices within 2 mm of correct position, confirmed by peer check. (2) During discussion, learner can state k numerically ($k = 2$ for small → medium, $k = 3$ for small → large) unprompted. (3) Learner's written sentence uses the word 'ratio' or 'times' rather than 'size' or 'bigger.' Teacher notes any learner who cannot place even one image vertex correctly — schedule one-on-one support before Lesson 3.

		us the RATIO of lengths, not the actual size. Say it again using the word ratio.'		
Driving Question Board (DQB) Creation  VIDEO: Dilations: center Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Corresponding Parts of Similar Figures (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	Each learner takes two sticky notes. On the GREEN note they write one thing they NOW KNOW that they did not know at the start of today's lesson (must include the word 'centre' or 'k'). On the ORANGE note they write one new question that today's lesson raised for them — ideally connecting to the driving question about the cups. Learners post both notes on the class DQB under the columns 'NOW WE KNOW' and 'WE STILL WONDER.' The class reads the orange notes aloud in a 2-minute gallery walk around the board.	Say: 'You have 3 minutes — individual, silent writing on your sticky notes. The green note must have the word CENTRE or K in it. The orange note must connect to our uji cups.' [Enforce 3-minute silence.] After posting, lead a 2-minute gallery walk: 'Walk around — read every orange question — put a small dot on any question that is ALSO in your head.' After the walk, identify the 2–3 most-dotted orange questions and read them aloud: 'These are our class's best wondering questions. Let's read them: [read]. Baraza ya darasa — class parliament — does anyone want to make a prediction about the answer? We will keep these questions up and see if we can answer them by Lesson 4.' Do NOT answer the questions now; validate them: 'These are exactly the kind of questions mathematicians ask.'	DQB as Living Document: the DQB is not a one-time poster but a running record of the class's evolving understanding. Requiring the word 'centre' or 'k' on the green note prevents vague entries ('I learned about enlargement') and ensures metacognitive specificity. The dot-voting on orange notes surfaces the most broadly felt uncertainties, which the teacher uses to sequence Lessons 3–5.	Observable indicators: (1) Green note contains a specific, accurate statement referencing 'centre of enlargement' or 'k = image/object.' (2) Orange note poses a question that goes beyond today's content (e.g., 'Does k affect how much water the cup holds?' or 'What happens to the area if k = 3?'). Teacher photographs the DQB after class to track conceptual progression across the unit.
Model Building Phase  VIDEO: Dilating shapes: shrinking <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Corresponding Parts of Similar Figures Practice	Learners open their 'Anchor Model' page — a running diagram they began in Lesson 1 showing the three uji cups side by side with their measurements. Today they ADD to the diagram: (a) they draw a dot labelled O above or to the left of the cups and draw rays from O through the top-rim of the small cup extending to the top-rim of the medium and large cups; (b) they label the scale factors $k = 2$ (small → medium) and $k = 3$ (small → large) on the rays; (c) they write in a box beside the diagram: 'Linear scale factor $k = \text{image length} \div \text{object length}$. For the uji cups: $k^2 = ???$ and $k^3 = ???$' — the	Say: 'Turn to your Anchor Model page. Today we are adding the mathematical engine — the centre and the scale factor. You have 8 minutes to update your diagram. The two ??? boxes should stay blank — they are your questions for Lesson 3.' [WAIT — 8 minutes working time.] Circulate and check: (a) Is O outside both cup shapes? (b) Do the rays pass through corresponding points on both cups? (c) Are $k = 2$ and $k = 3$ labelled on the correct rays? For any model where O is placed BETWEEN the cups, ask: 'If O is between the cups, which direction would you travel from O to reach	Cumulative Model Revision with Intentional Incompleteness: the ??? boxes are a deliberate instructional move — they make the model's incompleteness visible and motivating, directly sustaining the cognitive tension from the phenomenon. By requiring learners to draw O and the rays onto the cup diagram, the teacher ensures that the abstract geometric concept (centre of enlargement) is physically integrated into the anchoring phenomenon rather than existing as a separate textbook procedure.	Observable indicators: (1) O is placed outside both cup profiles, with rays correctly extending through corresponding points on each cup. (2) $k = 2$ and $k = 3$ are correctly labelled on the appropriate rays (small → medium and small → large respectively). (3) The two ??? boxes are present and blank — confirming the learner recognises the model is incomplete. Teacher collects 5 randomly selected Anchor Model pages to review before Lesson 3.

<i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	??? boxes are deliberately left blank to be filled in Lesson 3. Learners share their updated model with their partner and agree on the placement of O and the labels.	the small cup, and which direction to reach the large cup? Can that be right?' After 8 minutes: 'Show your partner your model. If your partner's O is in a different position from yours, discuss for 1 minute and agree on the correct position before we close.' Cold-call one pair to describe their model to the class in 30 seconds. Close with: 'Next lesson we will fill in those ??? boxes — and you will see why the answer is NOT 2 and NOT 3, and why that is the secret the uji cups have been hiding.'		
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D. TEACHER REFLECTION

After the lesson, consider: (1) How many learners successfully located the centre of enlargement independently (rays converging at one point) without teacher redirection? If fewer than 70% succeeded unaided, plan a 5-minute re-teach at the start of Lesson 3 using a different pair of similar shapes. (2) Which orange DQB questions most frequently appeared? If learners are already asking about area or volume, you can foreground those connections earlier in Lesson 3. (3) Did any learner's ratio table show significantly inconsistent values (e.g., ratios ranging from 1.8 to 3.1)? This signals a ruler-reading or measurement skill gap — address individually before the assessment task. (4) Were the uji cup connections made explicitly by learners, or only by the teacher? If learners are not yet volunteering the cup connection, increase the frequency of explicit anchoring prompts in Lessons 3 and 4.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners traced a pentagon ABCDE and its enlarged image on plain paper, drew rays through three pairs of corresponding vertices ($A \rightarrow A'$, $B \rightarrow B'$, $C \rightarrow C'$), and identified the single point O where all rays intersected as the centre of enlargement. They also measured five pairs of corresponding sides on the pentagon and computed the ratio image÷object for each pair.
What did I learn?	The centre of enlargement O is the unique fixed point from which every vertex of the object is projected outward to its corresponding image vertex. The linear scale factor $k = \text{image length} \div \text{object length}$ is constant for ALL pairs of corresponding lengths in an enlargement — it does not vary from side to side. For the uji cups, $k = 2$ (small → medium) and $k = 3$ (small → large), meaning every single length on the medium cup is exactly twice the matching length on the small cup.
How does this explain the phenomenon?	Because k is the ratio of ANY matching pair of lengths, the manufacturer can guarantee that the large uji cup is geometrically similar to the small one simply by using the same centre of enlargement and the same scale factor k for every point. This is why the cups 'look the same' despite being different sizes. However, k only describes lengths — the class has not yet explained why filling the cup requires k^2 times as much water in cross-section, or k^3 times as much in volume. Those ??? boxes in the Anchor Model will be resolved in Lesson 3.

LESSON 3 (40 minutes): Drawing Enlargements — Constructing Images from a Centre and Scale Factor

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners construct the image of a given figure using a centre of enlargement and a linear scale factor ($k = 2, 3$, and $\frac{1}{2}$), then verify by measurement that all lengths scale by k while all angles remain unchanged — directly connecting the uji cup k -pattern to a precise geometric construction procedure.
Knowledge	– The image of each vertex is found by drawing a ray from the centre of enlargement O through the object vertex and measuring k times the object distance along that ray. – Under enlargement by scale factor k, every length in the image is exactly k times the corresponding object length, but every angle remains equal to its corresponding object angle. – A scale factor $k > 1$ produces an enlargement (image larger than object); $0 < k < 1$ produces a reduction (image smaller than object); $k = 1$ leaves the figure unchanged.
Skills	– Accurately construct the image of a given polygon given a centre of enlargement and a linear scale factor using a ruler and straight edge. – Verify an enlargement construction by measuring corresponding lengths and confirming the ratio equals k, and by measuring corresponding angles and confirming they are equal.
Attitudes	– Appreciate the precision required in geometric construction and the satisfaction of a verified result. – Recognise that the mathematical pattern discovered in the uji cup phenomenon (scaling by k) is not accidental — it is a necessary consequence of how enlargement works.
Key Inquiry Question	How is similarity and enlargement applied in real life?
Purpose in Storyline	Lessons 1 and 2 established what similarity means and where the centre of enlargement is. This lesson translates that understanding into action: learners now construct enlargements themselves, confirm that every length scales by k while every angle is preserved, and begin to see WHY the uji cup areas came out as k^2 — because both dimensions of the cup's cross-section scaled by k simultaneously.
Safety Notes	Learners use rulers and pencils with sharp points. Remind learners to keep compass/pencil points directed downward and away from peers when passing materials. No other specific hazards.

B. LESSON OVERVIEW

In this third lesson of the evidence-gathering sequence, learners move from recognising and measuring similarity (Lessons 1–2) to actively constructing it. Each learner draws the image of a simple polygon — first a triangle, then a rectangle — given a printed centre of enlargement O and a stated linear scale factor. They work with $k = 2$, $k = 3$, and $k = \frac{1}{2}$ in turn, mirroring the exact ratios found on the uji cups (small → medium and small → large). After constructing each image, learners measure every pair of corresponding sides and confirm the ratio equals k , then use a protractor to confirm every pair of corresponding angles is equal. This measurement-verification step is the core evidence-gathering activity of the lesson.

The lesson is anchored throughout in Kenyan real-life contexts: a photographer in Mombasa who must enlarge a portrait to poster size for a political rally ($k = 3$), and a cartographer at the Survey of Kenya who scales a detailed street map of Kisumu down to fit a regional atlas page ($k = \frac{1}{2}$). These contexts frame the central question — what changes and what stays the same? — and lead learners to articulate the fundamental result: enlargement preserves shape (angles) but changes size (lengths).

This distinction is then connected back to the uji cups: the cups are enlargements of one another, so every length on the medium cup is exactly twice ($k = 2$) the matching length on the small cup, and this is why the cross-sectional area came out four times — both the diameter and the height scaled by 2 at the same time.

By the close of the lesson, learners update their Driving Question Board with two new claims: (1) construction gives us a precise tool for predicting where the image of any point will land, and (2) angles are invariant under enlargement, but lengths are not — they scale by k . These claims directly advance the driving question by explaining the mechanism behind the uji cup mystery and setting up Lesson 4's investigation of area scale factor (k^2).

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Dilations: scale factor Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Dilations: scale factor Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Each learner receives a half-sheet showing a small triangle ABC printed on plain paper, with a point O clearly marked 4 cm to the left of vertex A. Before touching a ruler, learners individually sketch — in pencil, on the same sheet — where they think the image triangle A'B'C' will appear if the figure is enlarged by $k = 2$ from centre O. They annotate their sketch with a sentence: 'I think the image will be [larger/smaller/same size] because...' and a second sentence predicting whether the angles will change. Learners then share predictions with a shoulder partner for 60 seconds.	Distribute the predict sheets face-down. Say: 'Do NOT turn over until I say go. You are going to predict — not calculate, not construct — just sketch your best guess.' After 90 seconds of individual silent sketching, say: 'Tell your partner two things: where you drew the image, and whether you think the angles will stay the same or change. You have 60 seconds — go.' After partner talk, cold-call 3 learners (choose one confident, one hesitant, one who looks uncertain): 'Akinyi, where did you draw your image — farther from O or closer?' — WAIT 10 seconds. 'Omondi, same or different angles?' — WAIT 10 seconds. 'Fatuma, do you agree with Omondi — why or why not?' — WAIT 12 seconds. Record the three prediction positions on the board as quick sketches labelled P1, P2, P3. Say: 'We will come back to these predictions at the end of the Observe phase and see who was closest — and more importantly, WHY.'	Prediction Sketching with Annotated Reasoning — learners externalise their mental model of enlargement before instruction. By committing to a sketch and a written sentence, they activate prior knowledge from Lessons 1–2 (centre of enlargement, k as a ratio) and create a cognitive anchor. The subsequent partner share surfaces misconceptions (e.g., 'the image just slides to the right' or 'angles get bigger too') that the teacher can target during the Observe phase.	Teacher scans the room during sketching and looks for: (1) learners who place the image on the opposite side of O from the object — a sign they are confusing enlargement with reflection; (2) learners who draw an image the same size as the object — indicating $k = 2$ has not been internalised; (3) learners who correctly place the image farther from O and larger. Teacher notes these three groups mentally to differentiate cold-call questions in the Observe phase.
Observe Phase  VIDEO: Dilating shapes: shrinking	Learners work in pairs to perform three timed construction tasks on pre-printed worksheets, each showing an object figure, a marked centre O, and a stated scale	Write the four-step construction protocol on the board before distributing worksheets. Read each step aloud and mime the ruler action. Say: 'Step 1 — ray	Structured Construction with Immediate Measurement Verification — the four-step protocol makes the abstract definition of enlargement (image	Teacher checks verification tables during circulation. Success indicators: (a) all three side-length ratios in the table equal k (within ± 0.1 cm measurement tolerance);

Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Dilations: scale factor Source: Khan Academy (English - US curriculum) Search ARES for similar readings	factor. Task 1: Enlarge triangle PQR by $k = 2$ from centre O (mimicking the small-to-medium uji cup). Task 2: Enlarge the same triangle by $k = 3$ from the same O (small-to-large cup). Task 3: Reduce rectangle ABCD by $k = \frac{1}{2}$ from O (mimicking the cartographer in Kisumu shrinking a map). For each task, learners use the step-by-step construction protocol written on the board: (a) Draw a ray from O through each vertex. (b) Measure the distance from O to the vertex. (c) Multiply by k and mark the image vertex on the ray at that distance. (d) Connect image vertices to form the image figure. After constructing, they fill in a verification table: measure each pair of corresponding sides and compute the ratio; measure each pair of corresponding angles and record.	from O through the vertex. Step 2 — measure O to vertex. Step 3 — multiply by k, mark the new point. Step 4 — join the new points. Simple. Let us do Task 1 together for just the first vertex P, then you continue.' Demonstrate on a large printed copy taped to the board: measure OP aloud ('I measure 3 cm'), multiply ('$3 \times 2 = 6$ cm'), mark P' ('I mark 6 cm from O along the ray'). Say: 'Now you do vertices Q and R and connect P'Q'R'. You have 4 minutes.' Circulate — at the 2-minute mark, cold-call one pair: 'Show me your ray for vertex Q — is Q' farther from O or closer than Q?' — WAIT 10 seconds. After Task 1, pause the class: 'Check your verification table — what ratio did you get for side P'Q' $\div$ PQ? Raise your hand if you got exactly 2.' Scan room. 'What about the angle at Q' compared to Q?' Cold-call 2 pairs. Repeat the pause-and-check after Tasks 2 and 3. Connect to context: 'This is exactly what a photographer in Mombasa does when blowing up a portrait to poster size — every length triples, but Uncle Kamau's face still looks like Uncle Kamau because the angles stay the same.' For Task 3, ask: 'The cartographer at Survey of Kenya uses $k = \frac{1}{2}$ — what happens to the distances from O?' — WAIT 12 seconds — cold-call 2 learners.	distance = $k \times$ object distance from O) into a concrete, repeatable physical procedure. The verification table forces learners to generate their own quantitative evidence that lengths scale by k and angles do not. This is the same evidence-gathering logic used in Lesson 1 when learners measured the uji cups: observe $\rightarrow$ measure $\rightarrow$ compute ratio $\rightarrow$ draw conclusion.	(b) all three angle measurements in the table show equality between object and image (within $\pm 2^\circ$). Red-flag indicators: any ratio that is not k suggests the learner measured from the vertex rather than from O; any angle discrepancy greater than 5° suggests incorrect image vertex placement. Teacher notes pairs with errors for targeted support during the Explain phase.
Explain Phase VIDEO: Dilating shapes: expanding Source: Khan Academy (English - US curriculum) Search ARES for similar videos	The class reconvenes. Each pair writes three statements on a sticky note (or speaks aloud if sticky notes are unavailable): (1) 'Under enlargement by k, lengths [increase/decrease/stay the same] by a factor of ____.' (2) 'Under enlargement by k, angles [increase/decrease/stay the	After collecting sticky notes, read out three contrasting responses (one correct, one partially correct, one with a misconception about angles). Do NOT say which is right yet — say: 'Three different ideas on the board — let us use our verification tables to decide.' Cold-call the pair whose verification	Claim-Evidence-Reasoning (CER) Protocol — learners construct explicit written claims ('lengths multiply by k'), identify the evidence from their verification tables (measured ratios), and begin to reason toward the k^2 result for area. This bridges the procedural skill of construction	Teacher listens to pair discussions during the 90-second think time. Success indicators: learners spontaneously use the phrase 'both dimensions scale by k' or write $\text{Area} = (kr)^2$. Misconception to watch for: learners saying 'area doubles because $k = 2$' — this shows they are adding k rather

 EXERCISE: Dilations: center <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	same].' (3) 'The centre of enlargement O [does/does not] change position.' Pairs share their statements. The teacher then poses the critical connection question: 'Think back to the uji cups. The medium cup had $k = 2$ compared to the small cup. Its height was $2\times$ and its diameter was $2\times$. If I want to know the area of the circular top of the medium cup compared to the small cup — I am scaling TWO lengths each by 2. What do you think happens to the area?' Learners discuss in pairs for 90 seconds, then the class works through the reasoning together: $\text{Area} = \pi \times r^2$. If r is doubled ($k = 2$), new area $= \pi \times (2r)^2 = 4\pi r^2 = 4 \times \text{original area}$. The teacher writes $k^2 = 4$ on the board and links it to the 4 small cups of water needed to fill the medium uji cup from Lesson 1.	table showed the clearest evidence: 'Read me your angle column for Task 1.' — WAIT 10 seconds. After confirmation, write the three conclusions on the board in learner-friendly language: 'LENGTHS: multiply by k. ANGLES: stay the same. CENTRE O: stays put.' Then say: 'Now here is the big connection. Pause and think: the uji cup phenomenon — why did the medium cup need FOUR small cups of water to fill the circular cross-section, not two? Talk to your partner. I will ask in 90 seconds.' — WAIT the full 90 seconds. Cold-call 3 pairs in sequence, building the argument incrementally: 'Wanjiku, what is the radius of the medium cup compared to the small one?' — WAIT 10 seconds. 'Kipchoge, so what is the area of the circular top of the medium cup?' — WAIT 12 seconds. 'Amina, does that match what we found in Lesson 1 with the water fills?' — WAIT 10 seconds. Write on the board: '$k = 2 \rightarrow k^2 = 4$ (area). $k = 3 \rightarrow k^2 = 9$ (area).' Circle both numbers and draw an arrow to the uji cup diagram on the wall.	(Observe phase) to the conceptual understanding of why areas scale as k^2. The teacher's Socratic cold-call sequence scaffolds the reasoning step-by-step rather than simply telling learners the answer.	than squaring it. If this misconception surfaces, the teacher draws a $1\text{ cm} \times 1\text{ cm}$ square, then a $2\text{ cm} \times 2\text{ cm}$ square, and asks: 'How many small squares fit inside the big one?'
Driving Question Board (DQB) Creation  VIDEO: Dilating shapes: shrinking by 1/2 Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Prove triangle similarity	Learners return to the class Driving Question Board (a large sheet of paper or section of the chalkboard divided into three columns: QUESTIONS WHAT WE KNOW WHAT WE THINK). Each pair writes one new 'WHAT WE KNOW' sticky note from today's lesson and one new 'QUESTION' that today's lesson has raised. Suggested sentence starters are displayed: 'We now know that...' and 'We still wonder...'. Pairs stick their notes in the correct columns. The	Give pairs 3 minutes to write their two sticky notes. While they write, circulate and prompt any pair that is stuck: 'What is the one thing you are sure of from the construction today?' and 'What is one thing you still do not understand about the uji cups?' After posting, read 4–5 KNOW notes and 3–4 QUESTION notes aloud. For each KNOW note, ask the class: 'Agree? Evidence?' — WAIT 8 seconds — quick show of hands. For each QUESTION note, say: 'Good question. Is this a question we can	Collaborative Knowledge Wall (DQB Update) — the Driving Question Board makes the class's collective understanding visible and dynamic. By distinguishing KNOW from QUESTION, learners practise epistemic humility (acknowledging what is not yet explained) while celebrating genuine progress. The teacher's live vote on each sticky note creates a low-stakes peer-review of claims, reinforcing that knowledge must be evidence-backed.	Teacher reads all KNOW notes quickly. Indicators of solid understanding: notes mention 'lengths multiply by k', 'angles stay equal', 'centre O fixed'. Indicators of confusion: notes that say 'the shape changes' or 'the image is always bigger' (forgetting $k < 1$ case). Teacher photographs the DQB at the end of the lesson for comparison with later lessons.

<i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar readings	teacher reads selected entries aloud and the class votes (thumbs up/sideways/down) on whether each belongs in KNOW or QUESTION.	answer today, or do we need more lessons?' Highlight with a star any question that anticipates Lesson 4 (area scale factor). Say: 'Look at this question — [read a k^2 question]. That is exactly where we are heading next lesson. Hold onto that thought.' Point to the uji cups on the demonstration table: 'We can now predict where the image of any point on a cup would land if we enlarged it. But we still cannot fully explain why doubling the height AND the diameter gives us four times the water. That is Lesson 4.'		
Model Building Phase  VIDEO: Dilations: scale factor Source: Khan Academy (English - US curriculum) Search ARES for similar videos  EXERCISE: Similarity & transformations <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar readings	Each learner opens their individual 'Similarity Model' — a cumulative diagram started in Lesson 1 showing the three uji cups side by side with annotated measurements. Today, learners add a new layer: they sketch a ray from a marked centre O through one key point on the small cup (e.g., the top rim), extend it to the medium cup and to the large cup, and annotate: 'O to rim (small) = d; O to rim (medium) = $2d$; O to rim (large) = $3d$.' They also add a two-column table to their model titled 'What Scales What Stays the Same' and fill in at least two rows for each column. Finally, learners write a one-sentence revision to their model summary at the bottom of the page, updating their answer to the driving question: 'Doubling the size of the uji cup means every length doubles, but the area of the circle doubles TWICE because...'	Say: 'Open your Similarity Model from Lesson 1. You are going to add the construction idea to it — show me that the uji cups are literally enlargements of each other from a centre point O.' Give learners 5 minutes to draw and annotate. Circulate and prompt with: 'Where would O be if the three cups are enlargements of each other? Does it matter exactly where O is, as long as the ratios are right?' — WAIT 10 seconds per learner before offering a hint. With 2 minutes remaining, say: 'Now write your one-sentence revision at the bottom. Use the words: k, lengths, angles, area.' Cold-call 2 learners to read their revision sentence aloud. After each one, ask the class: 'Does that sentence answer part of our driving question — why doubling the cup gives eight times the volume? What is still missing from the sentence?' — WAIT 12 seconds. Close the lesson by pointing to the DQB: 'Our model is growing. We can now construct any enlargement and predict lengths. Next lesson, we will prove	Cumulative Model Revision — the individual Similarity Model is a running artefact that grows across all 8 lessons. Each revision forces learners to integrate new knowledge with prior understanding rather than treating each lesson as isolated. The 'What Scales What Stays the Same' table builds the conceptual foundation for the k^2 and k^3 results in Lessons 4 and 5 by making the distinction between invariant (angles) and variant (lengths) properties explicit and visual.	Teacher collects or photographs 4–5 model sheets at the end of the lesson. Success indicators: the ray diagram correctly shows O to small-cup point = d, O to medium = $2d$, O to large = $3d$; the two-column table correctly lists 'lengths' under 'scales' and 'angles' under 'stays the same'; the revision sentence includes k and either 'k^2' or 'both dimensions scale by k'. A learner who writes 'area doubles when $k = 2$' has not yet internalised the k^2 result and needs a brief one-on-one follow-up at the start of Lesson 4.

		that areas grow as k^2 — and that will bring us one step closer to explaining the full uji cup mystery.'		
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D. TEACHER REFLECTION

- 1. During the Observe phase, did all learners correctly place image vertices on the ray FROM O (not on a parallel ray or a translated position)? If not, which specific step in the four-step protocol caused the error — and how will I reteach that step at the start of Lesson 4?
- 2. When I posed the connection question ('why did the medium uji cup need four small cups, not two?'), how many pairs reasoned to k^2 independently versus needing my Socratic prompting? What does this tell me about readiness for the formal area scale factor lesson?
- 3. Were the 'QUESTION' sticky notes on the DQB genuine mathematical questions (e.g., 'what happens to area when $k = 3$?') or surface-level procedural questions (e.g., 'how do I use a ruler more accurately?')? How can I nudge the class toward deeper conceptual questions next lesson?
- 4. Did the $k = \frac{1}{2}$ reduction task (Task 3, the Kisumu cartographer context) produce any additional misconceptions — for example, learners placing the image BETWEEN O and the object versus beyond the object? How prevalent was this error?
- 5. Looking at the revision sentences in the Model Building phase, how many learners used the phrase 'both dimensions scale by k ' or equivalent reasoning? Is the class ready to formalise the area scale factor formula in Lesson 4, or do I need a brief bridging activity at the start of that lesson?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners constructed enlargements of triangles and rectangles using three different scale factors ($k = 2$, $k = 3$, $k = \frac{1}{2}$) from a centre of enlargement O, then measured every pair of corresponding sides and angles to verify their constructions. They saw that the image is always farther from O (for $k > 1$) or closer to O (for $k < 1$), and that the verification table consistently showed the length ratio equal to k and the angles equal.
What did I learn?	Under enlargement by scale factor k : (1) every length in the image equals k times the corresponding object length; (2) every angle in the image equals the corresponding object angle — shape is preserved; (3) the centre of enlargement O remains fixed. A scale factor $k > 1$ enlarges; $0 < k < 1$ reduces. These results hold for ANY pair of corresponding lengths — not just selected sides.
How does this explain the phenomenon?	This explains part of the uji cup phenomenon: because the cups are enlargements of each other, every single length on the medium cup is exactly $k = 2$ times the matching length on the small cup. The circular cross-section has both its radius AND its height scaled by $k = 2$. Since area involves two length dimensions multiplied together, the area scales by $k \times k = k^2 = 4$ — which is exactly why four small cups of water were needed to fill the medium cup's cross-section, not two.

LESSON 4 (80 minutes (2 × 40-minute lessons)): Area Scale Factor: Why Does the Uji Cup Label Need Nine Times as Much Paper?

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners discover and formalise the relationship between the linear scale factor k and the area scale factor k^2 , using cut-out similar rectangles and squares, and apply this relationship to explain the anchoring phenomenon of the uji cups.
Knowledge	Area scale factor of similar figures = k^2 , where k is the linear scale factor; the cross-sectional area of a circle scales by k^2 when the radius is multiplied by k .
Skills	Measure dimensions of similar cut-out figures; compute ratios of corresponding areas; tabulate results for $k = 2, 3, 4$; identify and generalise the k^2 pattern; apply the pattern to explain and predict real-world scaling situations.
Attitudes	Appreciate the power of mathematical generalisation; develop curiosity about why nature and manufactured objects follow scale-factor rules; show responsibility by handling shared cut-out resources carefully and taking turns during group work.
Key Inquiry Question	How is similarity and enlargement applied in real life?
Purpose in Storyline	In Lessons 1–3 learners identified similar figures, located centres of enlargement, and measured linear scale factors from the uji cups. They were puzzled that the medium cup needed FOUR small cups of water to fill its cross-section, not two. Lesson 4 resolves that puzzle: by measuring the areas of similar cut-out figures for $k = 2, 3, 4$, learners discover that area scales by k^2 , not k . They return to the uji cups and explain the cross-sectional mystery. This k^2 relationship is recorded on the Driving Question Board as the second layer of the cumulative class model.
Safety Notes	Scissors are used during cut-out preparation (pre-cut by teacher if time is limited). Remind learners to pass scissors handle-first, keep them flat on the desk when not in use, and not to run with scissors. No other significant hazards.

B. LESSON OVERVIEW

Learners begin with a prediction: if a rectangle is enlarged by a linear scale factor of $k = 3$, how many times larger is its area? After recording predictions, they work in groups with pre-prepared sets of similar squares and rectangles (original and images for $k = 2, 3, 4$) cut from manila paper. They measure dimensions, compute areas, tabulate the area ratios, and discover the pattern area scale factor = k^2 . The teacher then returns to the anchoring phenomenon — the three uji cups — and learners use their new formula to explain why four small cups fill the cross-section of the medium cup and nine fill that of the large cup. The lesson closes with learners revising their cumulative DQB model to include the k^2 layer, and a brief exit task where they predict the area of a shamba (farm plot) image given the original area and the linear scale factor.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO:	Each learner receives a half-sheet showing a small rectangle labelled	Distribute the half-sheets face-down. Say: 'Before you turn the	Elicit-and-record prediction: surfacing prior conceptions before	Scan the sticky-note predictions. If more than 60% of learners predict

Scale factors and area Source: Khan Academy (English - US curriculum) Search ARES for similar videos  EXERCISE: Dilations: scale factor <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar readings	'Original' (4 cm × 3 cm) and is told its image has been enlarged with $k = 3$. Without measuring or calculating, learners individually write two predictions on a sticky note: (1) What are the dimensions of the image? (2) How many times bigger is the area of the image compared to the original? Learners stick their predictions on a class 'Prediction Wall' drawn on the chalkboard under two columns: 'Dimensions ×3' and 'Area ×?'.	sheet over, I want you to think only about what you already know from our last three lessons. We found that linear scale factor k tells us how many times each LENGTH grows. Now here is the big question — does AREA also grow by the same factor?' Allow 10 seconds of silent thinking (WAIT TIME). Then say: 'Turn your sheet over. Write your two predictions on the sticky note. You have 3 minutes — this is individual thinking time, no discussing yet.' After 3 minutes, cold-call 5 learners (mix of confident and hesitant) to read their area predictions aloud. Record the spread of responses on the board (common answers will be ×3, ×6, ×9). Do NOT correct yet. Say: 'We have three different predictions. By the end of today's lesson, you will know with certainty which is right — and more importantly, WHY.'	instruction creates cognitive dissonance that motivates the investigation. Recording the spread of predictions publicly gives every learner a personal stake in the outcome.	×3 (the most common misconception — assuming area scales linearly), note this and plan to return to it explicitly during the Explain phase. If most predict ×9, proceed but still require them to prove it empirically.
Observe Phase  VIDEO: Area and Perimeter of Squares and Rectangles Source: CK-12 Search ARES for similar videos  EXERCISE: Dilations: scale factor <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar readings	Groups of 4 receive an envelope containing sets of pre-cut similar figures: Set A — three squares with sides 2 cm, 4 cm, and 6 cm ($k = 2$ and $k = 3$ relative to the smallest); Set B — three rectangles with dimensions 3 cm × 2 cm, 6 cm × 4 cm, and 9 cm × 6 cm. Each group also receives a blank recording table with columns: Figure Length (cm) Width (cm) Area (cm²) Ratio of Areas (Image:Original) k k^2. Learners measure all dimensions with rulers, compute all areas, fill in the table, and compute the ratios. They then add a $k = 4$ row by drawing (on squared paper) a 4th rectangle scaled by $k = 4$ from the original 3 cm × 2 cm rectangle, measure it, and complete the row. After completing the table, each	Distribute envelopes and recording sheets. Say: 'Your job right now is to be measurement scientists — like surveyors mapping plots of land in the Rift Valley. Measure carefully to the nearest millimetre, calculate the areas, and fill in every cell of your table. Do not skip the k^2 column — compute it even if you do not yet know why it is there.' Circulate continuously. At the 8-minute mark, cold-call one group: 'Group 3, what area did you get for the 4 cm × 4 cm square?' Confirm or prompt correction. At the 14-minute mark, pause the class: 'Raise your hand if your Area Ratio for $k = 2$ is 4.' Count hands aloud. 'Raise your hand if your Area Ratio for $k = 3$ is 9.' Count hands. Say: 'Look at your k^2 column. What do you notice?'	Data-table pattern recognition: by computing both the area ratio AND k^2 side-by-side in the same table, learners are guided to notice the numerical equivalence themselves rather than being told. The graph converts the pattern into a visual curve (parabola), reinforcing that the relationship is non-linear.	Check recording tables for two indicators: (1) correct area computations (e.g., $2 \times 2 = 4$, $4 \times 4 = 16$, ratio = $4 = 2^2$; $3 \times 2 = 6$, $6 \times 4 = 24$, ratio = $4 = 2^2$); (2) the k^2 column matches the area ratio column in every row. Groups with mismatched columns have either a measurement error or an arithmetic error — redirect them to re-measure before moving on. If a group's graph shows a straight line rather than a curve, this is a red flag — prompt: 'What is the area ratio when $k=1$? When $k=2$? When $k=3$? Can a straight line pass through all three points?'

	group plots 'k' on the x-axis and 'Area Ratio' on the y-axis on a provided grid and connects the points.	Allow 15 seconds WAIT TIME. Cold-call 3 groups to share their observation. For the $k = 4$ extension row, circulate and check that learners are drawing the rectangle accurately on squared paper (12 cm $\times$ 8 cm) before they measure.		
Explain Phase  VIDEO: Scale factors and area Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Dilations: scale factor <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	The class reconvenes. Learners first revisit their sticky-note predictions on the Prediction Wall and individually mark their prediction as correct or incorrect. The teacher facilitates a whole-class discussion to formalise the rule: Area Scale Factor = k^2 . Learners then apply the rule directly to the anchoring phenomenon: given that the medium uji cup has $k = 2$ relative to the small cup, they calculate the expected cross-sectional area ratio ($k^2 = 4$) and confirm it matches the observed '4 small cups to fill the cross-section.' They repeat for the large cup ($k = 3$, $k^2 = 9$). Finally, each learner completes three 'Apply It' problems in their exercise books: (A) A square shamba in Kericho has side 10 m; its image on a map has linear scale factor $k = 5$ — find the area of the image shamba. (B) A rectangular billboard in Nairobi is similar to a small poster; the billboard is $k = 4$ times taller — how many times more paint does the billboard need? (C) Two similar triangular road signs — the larger has $k = 3$ relative to the smaller; the smaller has area 150 cm ² — find the area of the larger sign.	Open by returning to the Prediction Wall: 'Look at your sticky note. Now that you have the data, who predicted $\times 9$ for $k = 3$? Who predicted $\times 3$? Let us understand WHY $\times 3$ is wrong.' Draw a 1×1 square on the board. Say: 'If I scale every side by $k = 3$, I get a 3×3 square. The area was 1. The new area is 9. The area grew by $3 \times 3 = k \times k = k^2$.' Pause 10 seconds WAIT TIME. Ask: 'Why can we NOT just multiply the area by k ?' Cold-call 4 learners. Guide to the explanation: 'Because area has TWO dimensions — length AND width — and BOTH scale by k , so the total scaling is $k \times k$.' Then say: 'Now let us go back to our uji cups. The medium cup has $k = 2$. What does our formula predict for the cross-sectional area ratio?' WAIT 10 seconds. Cold-call 3 learners. Write on the board: $k^2 = 2^2 = 4$. Say: 'Does this match what we measured in Lesson 1? Yes — we needed 4 small cups!' Repeat for $k = 3$ (large cup). For the Apply It problems, give 8 minutes individual silent work, then cold-call one learner per problem to present their solution step-by-step at the board. Correct misconceptions immediately.	Claim-Evidence-Reasoning (CER): learners make a claim (area scales by k^2), cite evidence (their data table), and reason through the mechanism (two-dimensional scaling). Connecting the formula to the phenomenon closes the explanatory loop and deepens retention.	During Apply It, circulate and check exercise books. Look for: (A) correct answer 2500 m ² ($10 \times 5 = 50\text{m}$, $50^2 = 2500$ — note: this tests whether learners incorrectly apply k^2 to the side length rather than directly; the correct method is original area $\times k^2 = 100 \times 25 = 2500$ m ²); (B) 16 times more paint ($k^2 = 16$); (C) $150 \times 9 = 1350$ cm ² . A learner who gets $150 \times 3 = 450$ cm ² for (C) is still applying the linear scale factor to area — intervene with a targeted re-explanation using the 1×1 square diagram.
Driving Question Board (DQB) Creation	The class gathers around the shared DQB (a large poster or section of the chalkboard). The DQB currently shows the driving	Say: 'Take 3 minutes in your groups to agree on ONE thing you now know — write it on the green card — and ONE thing you still	Public knowledge-building on the DQB: making the class's growing understanding visible and cumulative reinforces	Read the green 'We now know' cards to check for accurate use of the k^2 formula. A card that says only 'area gets bigger' without

 VIDEO: Area and Perimeter of Squares and Rectangles Source: CK-12  Search ARES for similar videos  EXERCISE: Dilations: scale factor <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	question at the top, the uji cups phenomenon diagram, the linear scale factor definition from Lesson 2, and the centre-of-enlargement construction from Lesson 3. Each group writes one new 'We now know...' statement on a coloured card (green) and sticks it onto the DQB in the 'Area' section. They also write one remaining 'We still wonder...' question on an orange card — typically these will anticipate volume (e.g., 'If area scales by k^2, does volume scale by k^3?'). The teacher adds a large annotated arrow on the DQB connecting the k (linear) node to the new k^2 (area) node, labelled 'Lesson 4: Area Scale Factor.'	wonder — write it on the orange card. Use mathematics, not just words.' After 3 minutes, invite each group to place their cards. Read 2–3 orange 'wonder' cards aloud. Say: 'Look at these questions about volume. We have solved the cross-section mystery today. Next lesson we will investigate whether the volume mystery — why 9 small cups fill the large uji cup — follows the same kind of rule.' Draw the arrow on the DQB explicitly and label it. End by re-reading the driving question aloud: 'Why does doubling the size of a container make it hold eight times as much? We have now explained part of this — the area part. The volume part is coming.'	metacognition — learners see their own intellectual progress — and sets up productive anticipation for the next lesson.	referencing k^2 indicates incomplete understanding. A card that says 'area scale factor = k^2 because both length and width are multiplied by k' indicates strong understanding. Count the ratio of precise to vague cards as a class-level comprehension indicator.
Model Building Phase  VIDEO: Dilations: scale factor <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Dilations: scale factor <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Each learner opens their individual Model Building Journal (an exercise book maintained across all 8 lessons). In Lessons 1–3 they drew a diagram of the three uji cups with arrows showing the linear scale factor k. Today they add a new layer: they draw two similar rectangles (the original and an image with $k = 3$) side-by-side, label all dimensions, and annotate: 'Area of image = $k^2 \times$ Area of original = $9 \times$ Area of original.' They also add a speech bubble next to the medium uji cup: 'Cross-section area $\times 4$ because $k=2$ and $k^2=4$.' At the bottom of the page, learners write a one-sentence predictive model statement: 'I predict that if we scale all lengths of the uji cup by k, the volume will scale by ____ because ____.' (This intentionally foreshadows Lesson 5.)	Say: 'Open your Model Building Journal to the next blank page. You have 7 minutes to draw and annotate your updated model. The model must include: your similar rectangles with dimensions labelled, the formula Area Scale Factor = k^2, and the updated uji cup diagram. Finally, write your prediction sentence about volume — do not leave it blank, even if you are not sure. A good scientist makes a prediction and then tests it.' Circulate and give specific verbal feedback: 'I can see you have k^2 labelled — can you also show me WHERE the k^2 comes from geometrically, by drawing the small square nine times fitting inside the large square?' With 2 minutes remaining, cold-call 2 learners to share their prediction sentences. Record the two most common volume predictions on the board (likely k^2 and k^3) for reference at the start of Lesson 5.	Cumulative model revision: each lesson's model entry builds on the previous, creating a personal artefact that tracks conceptual growth. The forced prediction sentence activates forward-looking reasoning and creates the cognitive hook for the next lesson's investigation.	Collect Model Building Journals (or photograph the pages) at the end of the lesson. Check for: (1) correct labelling of dimensions and k^2 annotation; (2) the uji cup speech bubble correctly showing $\times 4$ for $k=2$; (3) a prediction sentence that attempts to generalise to volume (any prediction is acceptable — what matters is that learners are reasoning forward). Flag journals with no prediction sentence or with a drawing that confuses the image and original for targeted follow-up at the start of Lesson 5.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) What percentage of learners correctly predicted $\times k^2$ (rather than $\times k$) before the investigation — and did the balance shift by the end of the Explain phase? If more than 40% still showed the $\times k$ misconception in the Apply It problems, plan a 5-minute recap at the start of Lesson 5 using a concrete 'tile the floor' visual. (2) Did the data-table activity take longer than planned? If groups struggled with ruler measurement accuracy, consider pre-printing the cut-outs with dimensions already labelled for lower-readiness groups. (3) Which 'We still wonder' orange cards best capture genuine curiosity about volume? Use the most insightful one as the opening hook for Lesson 5. (4) Did the connection between the formula and the anchoring phenomenon feel genuine to learners, or did it feel forced? If learners did not spontaneously connect $k^2=4$ to '4 small cups,' plan a more explicit physical demonstration at the start of Lesson 5 (e.g., actually pouring water into circular cross-sections cut from card). (5) Were all learners able to access the Apply It problems, or did some learners need scaffolding on basic area computation before they could engage with the scale factor concept? Identify these learners for peer-support pairing in Lesson 5.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners measured and compared the areas of sets of similar squares and rectangles (with $k = 2, 3, 4$) cut from manila paper, recorded their data in a structured table, and plotted area ratio against k on a graph.
What did I learn?	The area scale factor of similar figures equals k^2 (the square of the linear scale factor), because both length and width are each multiplied by k , making the total area grow by $k \times k$. This explains why the cross-sectional area of the medium uji cup ($k = 2$) is 4 times — not 2 times — the cross-section of the small cup.
How does this explain the phenomenon?	Using the k^2 rule, learners explained and predicted how the cross-sectional area of a similar shape changes when the linear scale factor is known, and applied this to real-world contexts including shamba areas in Kericho, billboard paint coverage in Nairobi, and triangular road signs.

LESSON 5 (80 minutes (double lesson)): Volume Scale Factor: Why the Large Uji Cup Holds Eight Times as Much

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	By the end of this lesson, learners will be able to determine the volume scale factor of similar solids and explain why scaling all linear dimensions of a container by k multiplies its volume by k^3 .
Knowledge	Learners will know that: (i) volume scale factor = k^3 where k is the linear scale factor; (ii) this relationship holds for all similar solids (cubes, cuboids, cylinders, sufuria, uji cups); (iii) the three scale factors relate as: linear $k \rightarrow$ area $k^2 \rightarrow$ volume k^3 .
Skills	Learners will be able to: (i) measure the dimensions of similar solids and compute their volumes; (ii) tabulate and calculate the ratio of volumes for $k = 2$ and $k = 3$; (iii) use volume scale factor to predict the volume of any similar solid without re-measuring; (iv) explain the anchoring phenomenon fully using the k^3 rule.
Attitudes	Learners will appreciate that mathematical patterns discovered through hands-on measurement are powerful predictive tools, and will develop confidence in using those patterns to solve real-life design and packaging problems encountered in Kenyan contexts such as catering, pottery, and container manufacturing.
Key Inquiry Question	How is similarity and enlargement applied in real life?
Purpose in Storyline	In Lessons 3–4 learners established that area scales by k^2 . Now in Lesson 5 they extend that reasoning one dimension further: they physically measure and compute the volumes of similar Kasuku tins and model cubes/cuboids, discover the k^3 pattern, and for the first time can fully explain the anchoring phenomenon — why the large uji cup needs nine small cups of water ($k = 3$, so volume scale factor = $3^3 = 27 \div 3 = 9$ cross-section portions, and fully 27 times the volume). The DQB is updated with k^3 and the complete chain $k \rightarrow k^2 \rightarrow k^3$ is now visible.
Safety Notes	Learners handle water and tins — wipe spills immediately to prevent slipping. No sharp cutting tools are used in this lesson. Learners should handle Kasuku tins carefully to avoid sharp edges on cut lids.

B. LESSON OVERVIEW

Learners begin by revisiting the uji cup phenomenon and predicting how many small cups will fill the large cup when $k = 3$. They then measure the dimensions of two sets of similar solids (model cubes/cuboids OR small and large Kasuku tins) and compute volumes, tabulating the volume ratio for $k = 2$ and $k = 3$. Through structured discussion (Think-Pair-Share and data pooling on the board), they discover that volume scale factor = k^3 . They apply this rule to explain the anchoring phenomenon fully and connect it to the sufuria supporting phenomenon. The DQB is updated to record the complete scale-factor chain. A model-building activity asks learners to revise their cumulative diagrams to include the k^3 relationship. Formative assessment is embedded throughout via whiteboards, cold-calling, and a 3-question exit ticket.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Area and Volume	Learners are shown the three uji cups from the anchoring phenomenon (or a	1. Display the uji cup diagram/photograph prominently. Say: 'Look at the small cup and the	Individual Prediction with Public Commitment: By writing predictions before any instruction	Observable indicator: All learners have a numerical prediction on their whiteboard within 2 minutes.

of Similar Solids Principles Source: CK-12 Search ARES for similar videos  HTML: Area and Volume of Similar Solids (Flexbook) Source: CK-12 Search ARES for similar readings	diagram/photograph of them on the board). They are reminded that in Lesson 4 they proved the label (surface area) of the large cup ($k = 3$) needs 9 times as much paper as the small cup. The teacher now poses: 'We have never directly measured how many small cups of water fill the large cup. Based on everything you now know about k and k^2, what is YOUR prediction for the volume ratio when $k = 3$? Write your prediction AND your reasoning on your mini-whiteboard. Do not discuss yet.' Learners write individual predictions (expected range: some say $3\times$, some say $9\times$, some who are beginning to generalise say $27\times$). After 2 minutes of silent writing, the teacher calls on 5 learners cold-call (by name from the register, not volunteers) to share predictions and reasoning. The teacher records all unique predictions on the board under the heading 'Our Predictions — Volume Ratio for $k = 3$' without affirming or correcting any.	large cup. Every length on the large cup is exactly 3 times the corresponding length on the small cup — that is our $k = 3$.' WAIT 5 seconds. 'In Lesson 4 we proved the surface area ratio is 9 — that is $k^2 = 3^2 = 9$. Today we go one step further: VOLUME. How many small cups of water do you think will fill the large cup?' 2. Say: 'Write your numerical prediction AND at least one sentence of reasoning on your whiteboard. You have 2 minutes. Work silently.' Start timer visibly. 3. After 2 minutes, cold-call exactly 5 learners: 'Amina — share your prediction and reasoning.' WAIT 10 seconds after each learner speaks before responding or moving on. 4. Record all predictions on the board without evaluation: e.g. '$3\times$', '$6\times$', '$9\times$', '$27\times$'. Say: 'We have several different predictions. Today's experiment will tell us who is right — and more importantly, WHY.'	and having them recorded publicly, learners create cognitive stakes in the outcome. This activates prior knowledge from Lessons 3–4 (k and k^2) and primes them to look for the pattern that resolves the discrepancy between predictions.	Teacher scans boards while circulating — note any learners who write '27' with correct reasoning (k^3 anticipated) and any who write '3' (still holding linear intuition). This informs pacing: if more than half predict 27 correctly with reasoning, compress Phase 2 to 15 minutes and extend Phase 3 discussion.
Observe Phase  VIDEO: Area and Volume of Similar Solids Principles Source: CK-12 Search ARES for similar videos  EXERCISE: Area and Volume of Similar Solids Practice Source: CK-12 Search ARES for similar readings	Learners work in groups of 4. Each group receives: (a) a set of 3 similar wooden or cardboard model cubes with side lengths approximately 2 cm, 4 cm, and 6 cm ($k = 1, 2, 3$ relative to the smallest), OR (b) a small Kasuku tin and a large Kasuku tin whose diameter and height are in ratio 1:2. Groups also receive a 30 cm ruler, a recording worksheet, and a calculator. PART A — Measurement and Computation (15 min): Learners measure the length, width, and height (or radius and height for cylinders) of each solid to the nearest millimetre and record in the worksheet table.	1. Distribute materials. Say: 'Your task has two parts. Part A: measure, compute volumes, and find the ratio. Part B: add your results to the class table. You have 15 minutes for Part A.' Display the worksheet structure on the board. 2. Circulate actively. Stop at each group for at least 60 seconds. Use targeted prompts: 'How did you decide which formula to use?' WAIT 10 seconds. 'What units will your volume be in?' WAIT 10 seconds. 'Before you divide — what do you expect the ratio to be based on our predictions?' 3. After 15 min, say: 'One spokesperson per group — write your k, your	Data Pooling + Cross-Group Comparison: By pooling data from groups using different solid shapes (cubes, cuboids, cylinders), learners see that the k^3 rule is shape-independent. This is a deliberate generalisation move — the pattern is more convincing when it appears across different physical objects than when seen in a single case.	Observable indicators: (1) Each group's worksheet table is complete with correct volume calculations (teacher checks units — cm^3 — and formula use during circulation). (2) Volume ratios in the class table are approximately 8 (for $k = 2$) and 27 (for $k = 3$) — flag any group with a ratio far from these for immediate re-measurement support. (3) Whiteboard sentences explicitly mention 'k^3' or 'cube of k' — learners who write only 'it is bigger' need a targeted question: 'By exactly how much? Compare the ratio number to k and to k^2.'

	They then compute the volume of each solid using the appropriate formula ($V = l \times w \times h$ for cuboids; $V = \pi r^2 h$ for cylinders). They calculate the ratio $\text{Volume}_2 / \text{Volume}_1$ (large $\div$ small) for $k = 2$ and $\text{Volume}_3 / \text{Volume}_1$ (largest $\div$ small) for $k = 3$, and record these ratios as decimals and as fractions. PART B — Class Data Pool (8 min): One representative from each group writes their group's results on the shared class table on the board (pre-drawn with columns: Group k Measured volume ratio k^2 k^3). All groups' data is now visible. PART C — Pattern Identification (5 min): Learners examine the class table and write ONE sentence on their whiteboard answering: 'What do you notice about the volume ratio column and the k^3 column?'	measured volume ratio, k^2, and k^3 in the class table. You have 3 minutes.' 4. Once the table is filled, give learners 2 minutes to examine it silently. Then say: 'Write ONE sentence on your whiteboard: what pattern do you see between the volume ratio column and the k^3 column?' WAIT 15 seconds of full silence. 5. Cold-call 4 learners (choose at least one from a group that used cylinders and one from a group that used cuboids): 'Kamau, read your sentence.' WAIT 10 seconds after each. 6. Affirm the pattern: 'Every group found that the volume ratio equals k^3 — even though some of you used cubes, some used cuboids, and some used Kasuku tins. The shape does not matter. The rule is: volume scale factor = k^3.'		
Explain Phase  VIDEO: Dilations: scale factor <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Area and Volume of Similar Solids (Flexbook) Source: CK-12  Search ARES for similar readings	PART A — Conceptual Explanation (8 min): The teacher leads a whole-class structured explanation of WHY volume scales by k^3, using a visual decomposition on the board. For a cube of side 1 unit: $V = 1^3 = 1$. Scale by $k = 2$: each side becomes 2, $V = 2^3 = 8$. The teacher shows that you can fit exactly 8 copies of the small cube inside the large cube by drawing a $2 \times 2 \times 2$ grid inside the larger cube. Learners copy the diagram into their notebooks and annotate it for $k = 3$ (showing a $3 \times 3 \times 3 = 27$ grid). PART B — Phenomenon Explanation (7 min): Learners work individually for 4 minutes to write a full explanation (3–5 sentences) answering: 'Using the k^3 rule, explain exactly why the large uji cup ($k = 3$) needs 27 small cups of water to fill it completely — not 3 and not 9.' They then share with a	1. Draw a $1 \times 1 \times 1$ cube on the board. Say: 'Volume = $1^3 = 1$ cubic unit. Now I scale every side by $k = 2$.' Draw the $2 \times 2 \times 2$ cube beside it. 'Volume = $2^3 = 8$. But WHY 8? Watch.' Draw grid lines inside the large cube showing 2 layers, 2 rows, 2 columns. 'I can fit exactly 8 small cubes inside. One dimension gives me $k = 2$ copies. Two dimensions give me $k^2 = 4$. Three dimensions give me $k^3 = 8$.' 2. Say: 'Now YOU draw this for $k = 3$ in your notebook — show the $3 \times 3 \times 3$ grid. You have 3 minutes.' WAIT. Cold-call one learner to the board to draw their version. 3. Say: 'Now connect this to our uji cups. Write 3–5 sentences explaining why the large uji cup needs 27 small cups — not 3, not 9. Use the words: linear scale factor, volume scale factor, k^3. You have 4 minutes.' WAIT. 4. Pair-share: 'Share with	Visual Decomposition + Phenomenon Re-Explanation: The 3D grid diagram makes the abstract k^3 formula concrete and countable. Requiring learners to write the phenomenon explanation in their own words (then stress-test it with a partner) ensures they are not just reciting a formula but can deploy it as an explanatory tool. The sufuria transfer problem checks whether they can apply k^3 to a new Kenyan context.	Observable indicators: (1) Notebooks show a correctly drawn $3 \times 3 \times 3$ grid with 27 small cubes visible (teacher circulates and stamps or ticks correct diagrams). (2) Written phenomenon explanations include the words '$k = 3$', '$k^3 = 27$', and reference to three spatial dimensions — flag any explanation that says '$9 \times 3 = 27$' without explaining WHY multiplication occurs in a third dimension. (3) Sufuria whiteboard answers show '$k = 2$, $2^3 = 8$' — any learner who writes '$2^2 = 4$' is still conflating area and volume scale factors and needs targeted re-teaching in Phase 4 or as homework.

	partner (Think-Pair-Share, 3 minutes). PART C — Sufuria Connection (5 min): The teacher poses: 'Mama Wanjiku has a small sufuria and a large sufuria. The large sufuria has a diameter twice as big as the small one AND is twice as tall. She wants to cook ugali for a wedding. How many times more maize flour does she need compared to using the small sufuria?' Learners solve individually on whiteboards (answer: $k = 2$, volume scale factor $= 2^3 = 8$ — she needs 8 times as much flour). Cold-call 3 learners to share and justify.	your partner. One of you reads, the other checks: did they mention $k = 3$, $k^3 = 27$, and connect it to the three dimensions of the cup?' 3 minutes. 5. Cold-call 2 pairs to share their explanation. After each, ask the class: 'Thumbs up if their explanation mentioned all three: k, k^3, and the three dimensions of the cup.' 6. Sufuria problem: Read aloud slowly. Say: 'On your whiteboard — what is k? What is the volume scale factor? How many times more flour?' WAIT 15 seconds. Cold-call 3 learners. Confirm: '$k = 2$, volume scale factor $= 2^3 = 8$. Mama Wanjiku needs 8 times more flour. This is exactly the same mathematics as our uji cup — just a different container.'		
Driving Question Board (DQB) Creation  VIDEO: Measuring volume as area times length <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Area and Volume of Similar Solids Practice Source: CK-12  Search ARES for similar readings	The teacher directs learners' attention to the class DQB (a large chart or section of the board that has been accumulating across all 8 lessons). Learners review what is already posted: Lesson 1 — phenomenon observations and initial questions; Lesson 2 — properties of similar figures and linear scale factor k; Lesson 3 — linear scale factor k confirmed with measurements; Lesson 4 — area scale factor k^2. Now the class adds the new understanding from Lesson 5. Each learner writes on a sticky note (or a strip of paper): one CLAIM (something they now know for certain) and one QUESTION (something they are still wondering about or want to explore). Claims are posted under the green 'We Now Know' column; questions under the yellow 'We Are Still Wondering' column. The teacher then draws attention to the complete scale-factor chain now	1. Point to the DQB. Say: 'Let's look at where we started and how far we have come.' Read aloud the original phenomenon question posted in Lesson 1. 2. Walk through each lesson's addition briefly: 'Lesson 2 — we found k. Lesson 3 — we confirmed k with measurements. Lesson 4 — we discovered k^2. Today — we discovered k^3.' As you name each, point to the relevant sticky notes or entries. 3. Distribute sticky notes. Say: 'Write ONE claim — something you are now certain is true — and ONE question you still have. You have 3 minutes. Write in full sentences.' WAIT. 4. Invite 4 learners (cold-call) to read their claim aloud before posting it. After each, ask: 'Class — do we agree this is a proven claim? Thumbs up or thumbs sideways.' 5. Post all notes. Then draw a large arrow on the board connecting $k \rightarrow k^2 \rightarrow k^3$. Say: 'This chain is the	DQB Claim-and-Question Protocol: Distinguishing 'what we know' from 'what we are still wondering' develops metacognitive awareness of the boundary between evidence and inference. Publicly posting the $k \rightarrow k^2 \rightarrow k^3$ chain gives learners a visual organiser of the unit's conceptual arc and positions them to anticipate future lessons.	Observable indicators: (1) Claims are specific and evidence-based (e.g. 'Volume scale factor $= k^3$, proven by our Kasuku tin measurements') rather than vague (e.g. 'big things are bigger'). (2) Questions are mathematically productive (e.g. 'Does k^3 work for spheres too?') rather than procedural (e.g. 'How do you calculate volume?'). Teacher photographs the DQB at the end of this phase for record-keeping and to share on the ARES platform.

	visible on the DQB: $k \rightarrow k^2 \rightarrow k^3$, and asks learners to suggest what real-life design questions this chain might help answer (expected: 'How do engineers design containers of different sizes?', 'How do architects scale up buildings?', 'Why does a large sufuria cost so much more to make than a small one?').	mathematical heart of our unit. In Lessons 6 and 7 we will use it to solve real design problems. In Lesson 8 we will bring it all back to our uji cups one final time.' 6. Read 2–3 of the 'still wondering' questions aloud. Say: 'These are excellent mathematical questions. Keep them in mind as we continue.'		
Model Building Phase  VIDEO: Volume word problem: water tank <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos  EXERCISE: Dilations: scale factor <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar readings	Learners retrieve their cumulative model diagram begun in Lesson 1 and revised in Lessons 2 to 4 (a labelled drawing or concept map showing a small solid and a large similar solid with annotations). They now add the volume scale factor layer: they draw a third annotation arrow on their diagram labelled 'Volume scale factor = k^3 ' and add a worked numerical example for $k = 3$ (showing $3^3 = 27$). They also add a brief 'phenomenon connection' box: 'Large uji cup ($k = 3$): volume = $27 \times$ small cup. Needs 27 small cups of water.' Learners who finish early extend their model to include the complete chain $k \rightarrow k^2 \rightarrow k^3$ in a table format with columns for $k = 1, 2, 3, 4$. The lesson closes with a 3-question exit ticket on mini-whiteboards: Q1 — 'A small cube has side 3 cm. A similar cube has side 9 cm. What is k ?' (Answer: 3). Q2 — 'What is the volume scale factor?' (Answer: 27). Q3 — 'If the small cube holds 27 cm^3 of water, how much does the large cube hold?' (Answer: 729 cm^3). Learners hold up whiteboards simultaneously on the teacher's signal.	1. Say: 'Take out your cumulative model diagram. You are going to add today's discovery.' Display the previous lesson's model structure on the board as a reference. 2. Say: 'Add a third arrow for volume scale factor = k^3 . Add a worked example for $k = 3$. Add a phenomenon connection box. You have 6 minutes.' WAIT. Circulate. Prompt any learner who has not yet connected the model to the uji cups: 'Where on your diagram can you show that the large uji cup holds 27 times as much as the small cup?' 3. After 6 minutes say: 'Pause your diagrams. Exit ticket — three quick questions on your whiteboards.' Read Q1 aloud. 'Write only your answer — you have 20 seconds.' WAIT 20 seconds. 'Boards up.' Scan all boards. Affirm or correct briefly. Repeat for Q2 and Q3. 4. Close: 'Today we completed the scale-factor chain: k for lengths, k^2 for areas, k^3 for volumes. We can now fully explain our uji cup mystery. In Lesson 6 we will use all three scale factors to solve real design problems — including designing containers for a Nairobi catering company. Well done.'	Cumulative Model Revision: Requiring learners to physically add to a diagram they have been building across five lessons externalises the growth of their understanding. The layered model (k , then k^2 , then k^3) makes the mathematical structure visible as a coherent system, not three isolated facts. The simultaneous whiteboard exit ticket gives the teacher a rapid whole-class snapshot of mastery before the lesson ends.	Observable indicators: (1) Cumulative models show three correctly labelled annotation arrows (k , k^2 , k^3) with numerical examples — models missing any layer indicate incomplete integration of prior lessons. (2) Exit ticket Q1: all boards show '3' — any board showing '6' or '1/3' indicates confusion about direction of scale factor. (3) Q2: boards show '27' — any board showing '9' indicates conflation of area and volume scale factors (priority re-teaching target). (4) Q3: boards show '243' — any board showing '243' (multiplied by 9 not 27) confirms the k^2 vs k^3 confusion. Teacher records names of learners with incorrect Q2 or Q3 for follow-up in Lesson 6 opener.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) Did the class data pool in Phase 2 show consistent k^3 ratios across all groups, or did measurement errors obscure the pattern for some groups? If ratios were inconsistent, consider whether the model cubes/Kasuku tins were sufficiently precise — pre-measuring and marking the solids before the lesson may help in future iterations. (2) In Phase 3, how many learners were still conflating area and volume scale factors (writing 9 instead of 27 on the sufuria problem)? If more than 25% of learners made this error, begin Lesson 6 with a 5-minute retrieval activity comparing k , k^2 , and k^3 side by side before moving to applications. (3) Were the DQB 'still wondering' questions mathematically productive? If learners are not yet asking questions about new shapes (spheres, cones) or real-life design, the phenomenon connection may not yet be deep enough — consider assigning a home observation task: 'Find two similar containers at home and measure their dimensions. What do you predict the volume ratio will be?' (4) Were all five phases completed within 80 minutes? The most common time pressure is Phase 2 (measurement) — if groups are slow, pre-fill one row of the class table as a model before the lesson begins. (5) Note the names of learners flagged in exit ticket Q2 and Q3 and ensure they are seated near a strong peer explainer in Lesson 6 group work.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners measured the dimensions of similar solids (model cubes/cuboids or Kasuku tins with $k = 2$ and $k = 3$) and computed their volumes. They pooled class data in a shared table showing the measured volume ratio alongside k^2 , and k^3 for each scale factor.
What did I learn?	The volume scale factor of similar solids equals k^3 , where k is the linear scale factor. This means that when every length is scaled by $k = 3$, the volume becomes $3^3 = 27$ times as large. The complete chain is now established: length scales by k , area by k^2 , and volume by k^3 .
How does this explain the phenomenon?	Learners explained the anchoring phenomenon fully: the large uji cup ($k = 3$) needs 27 small cups of water — not 3 and not 9 — because volume is a three-dimensional quantity and each of the three dimensions is multiplied by k , giving $k \times k \times k = k^3$. They also applied this to the sufuria supporting phenomenon, confirming that a sufuria with $k = 2$ holds $2^3 = 8$ times as much ugali porridge as the small sufuria.

LESSON 6 (80 minutes): Relating Linear, Area and Volume Scale Factors — The Unified k , k^2 , k^3 Model

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners consolidate the three scaling relationships (k , k^2 , k^3) into a single unified mathematical model, solve mixed real-life problems involving similar solids, and build a complete annotated diagram-and-formula summary that advances the class Driving Question Board.
Knowledge	Learners know that for two similar figures or solids with linear scale factor k : corresponding lengths scale by k , corresponding surface areas scale by k^2 , and corresponding volumes scale by k^3 . They can state and apply all three relationships simultaneously from a single given value of k .
Skills	Learners can: (a) construct and interpret a unified $k / k^2 / k^3$ table for any given linear scale factor; (b) solve mixed problems requiring calculation of paint needed (area) and water held (volume) for similar tanks; (c) use digital devices to verify scale-factor calculations; (d) produce a fully annotated model (diagram + formula summary) in their notebooks and on the class DQB.
Attitudes	Learners appreciate the power and elegance of a single number k encoding three different scaling behaviours, and develop confidence in applying this unified model to predict real-world quantities such as paint, water capacity, and material costs for structures across Kenya.
Key Inquiry Question	How is similarity and enlargement applied in real life?
Purpose in Storyline	Lessons 1–5 established the phenomenon (the Shrinking Uji Cup), identified properties of similar figures, introduced the linear scale factor, and explored area and volume scale factors separately. Lesson 6 is the pivotal consolidation lesson: learners tie all three relationships together into one coherent model. Using the Rift Valley water-tank context, they solve problems that require choosing the correct scale factor (k , k^2 , or k^3) for a given physical quantity, verify answers with digital devices, and update both their notebook models and the class DQB with the complete formula summary — directly answering the driving question for the first time with full mathematical precision.
Safety Notes	No hazardous materials are used. If learners bring physical containers for demonstration, ensure surfaces are dry to prevent slipping. Supervise responsible use of digital devices and enforce the school's online-safety guidelines when accessing the Kenya Education Cloud or any online simulation.

B. LESSON OVERVIEW

Lesson 6 (80 min) is the consolidation and model-building apex of the Similarity and Enlargement sequence. The teacher opens by projecting an image of two real water tanks from the Rift Valley (or a hand-drawn diagram) and poses a mixed-context challenge: given a linear scale factor of 3, how many times more paint does the larger tank need, and how many times more water does it hold? Learners first predict using prior knowledge, then gather evidence by constructing a unified $k / k^2 / k^3$ table and solving a structured set of mixed problems — some giving k and asking for area or volume ratios, others working backwards from area or volume ratios to find k . Digital devices are used to verify all calculations and to explore an interactive simulation on the Kenya Education Cloud. A whole-class sensemaking discussion makes the unified model explicit. Learners then build a complete annotated model (labelled diagram of the uji cups + formula box) in their notebooks, and two volunteers update the class DQB with the finished formula summary. The lesson closes with an exit-ticket problem that requires learners to apply all three relationships simultaneously.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
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Predict Phase  VIDEO: Area and Volume of Similar Solids Principles Source: CK-12  Search ARES for similar videos  HTML: Area and Volume of Similar Solids (Flexbook) Source: CK-12  Search ARES for similar readings	Learners examine a projected image of two similar cylindrical water tanks labelled 'Nakuru Small Tank' and 'Nakuru Large Tank'. The teacher reveals that every linear dimension of the large tank is exactly 3 times that of the small tank ($k = 3$). Without any calculation, learners individually write three predictions in their notebooks: (i) How many times more metal sheet (surface area) is needed to build the large tank? (ii) How many times more water does the large tank hold? (iii) If it costs Ksh 4,500 to paint the small tank, how much will it cost to paint the large tank? Learners share predictions with a shoulder partner for 2 minutes, then selected pairs report to the class. The teacher records all predictions (not corrected yet) on the whiteboard under the heading 'Our Predictions — Lesson 6'.	Display the Rift Valley tank image and say: 'Before we calculate anything, I want your honest mathematical instincts. Write your three predictions — you have 3 minutes of silent thinking time.' [WAIT 3 minutes — do not prompt.] After pair-share, cold-call 4 pairs (select one high-confidence, one low-confidence, one who predicted k^2 correctly, one who predicted k^3 correctly) and record all responses verbatim. Say: 'Notice we have different answers on the board. By the end of today's lesson, every one of you will be able to prove which prediction is correct — and explain exactly why.' Circle the predictions but do not reveal answers. Explicitly connect to the phenomenon: 'Remember our uji cups — we were surprised that doubling the height gave us four times the water in cross-section. Today we are going to build the complete mathematical story for that surprise.'	Elicit-and-Record Prediction Wall — recording unresolved predictions publicly creates cognitive dissonance that motivates the evidence-gathering phase and gives learners a concrete benchmark to revise against during model-building.	Observable indicator: Each learner has written three distinct numerical predictions in their notebook before pair-share begins. Teacher scans notebooks during circulation to check that predictions are numerical (not just 'more' or 'less'). Note which learners predict k^2 for area and k^3 for volume (showing carry-forward understanding from Lessons 4–5) versus those who predict k for all three (showing the misconception to target in the Explain phase).
Observe Phase  VIDEO: Area and Volume of Similar Solids Principles Source: CK-12  Search ARES for similar videos  EXERCISE: Area and Volume of Similar Solids Practice Source: CK-12  Search ARES for similar readings	Learners work in groups of 4 to complete a structured Unified Scale-Factor Table activity. Each group receives (or draws in their notebook) a table with columns: [Linear Scale Factor k Area Scale Factor Volume Scale Factor Example: length of edge Example: surface area Example: volume]. Groups complete the table for $k = 2, 3, 4$, and 5, using their knowledge from Lessons 4 and 5. They then open a verified digital simulation on the Kenya Education Cloud (or teacher-provided GeoGebra file) on their learner digital devices and manipulate a 3-D cylinder — increasing its radius and height by	Distribute or project the blank Unified Scale-Factor Table template. Say: 'You have 12 minutes to complete columns 2 and 3 using what you already know from Lessons 4 and 5, then verify every entry using your digital device. If your table value and the app value disagree, that is important data — write down both and flag it for our discussion.' [Set a visible 12-minute timer.] Circulate and use targeted prompts: for groups stalling on $k = 4$, ask 'What is 4 squared? What is 4 cubed? How does that connect to your table?' [WAIT 10 seconds for response before elaborating.] After 12 minutes, pause the class:	Data-Table + Digital Verification — completing the table by hand first activates prior knowledge and forces explicit calculation, while the app verification provides immediate corrective feedback and reinforces the pattern through a second representational mode (dynamic 3-D model), deepening understanding of the $k / k^2 / k^3$ relationships.	Observable indicator: Every group's table has correct entries for at least $k = 2$ and $k = 3$ (area scale factors 4 and 9; volume scale factors 8 and 27) before the Explain phase begins. Teacher checks by asking each group to hold up their table and scanning for the $k = 3$ row. Groups with incorrect entries are flagged for targeted support during the Explain phase.

	factors of 2, 3, 4, and 5 — reading off the surface area and volume displayed by the app. They record the app's computed values alongside their table entries and check for agreement. Groups that finish early attempt the extension: $k = 1.5$ (representing a mug that is 1.5 times the height of the toddler uji cup).	'Hands up — which groups found a disagreement between their table and the app?' Address any disagreements publicly. Cold-call 3 groups to read out their $k = 3$ row. Say: 'Look at the row for $k = 3$ on your table. Freeze that row in your mind — it is the key row for our tank problem.'		
Explain Phase  VIDEO: Dilations: scale factor <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Area and Volume of Similar Solids (Flexbook) Source: CK-12  Search ARES for similar readings	The teacher leads a whole-class worked-example sequence using the Rift Valley tank scenario as the anchor. Three connected problems are solved together on the board, with learners solving each step in their notebooks before the teacher reveals the answer: Problem A (forward, area): 'Two similar water tanks in the Rift Valley have a linear scale factor of 3. A painter charges Ksh 500 per square metre. If the small tank has a surface area of 18 m^2, how much will it cost to paint the large tank?' Problem B (forward, volume): 'The small tank holds 2,000 litres of water. How many litres does the large tank hold?' Problem C (inverse): 'A third tank has a surface area 25 times that of the small tank. What is the linear scale factor between the small tank and this third tank? How many times more water does it hold?' After each problem, learners compare their notebook answer with the board solution and mark it. The teacher then formally states the Unified Model: 'If two similar figures have linear scale factor k, then area scale factor = k^2 and volume scale factor = k^3. One number — k — tells you everything.' Learners copy the boxed formula summary into their notebooks. A brief pair discussion	Write Problem A on the board and say: 'Work this out individually and silently — 3 minutes.' [WAIT 3 minutes.] Then say: 'Tell me: what scale factor do we use for surface area — is it k, k^2, or k^3? Show me with your fingers: one finger for k, two fingers for k^2, three fingers for k^3.' [WAIT 10 seconds for whole-class finger-vote.] Cold-call 2 learners who show different answers to explain their reasoning before revealing the solution. Repeat the finger-vote protocol for Problems B and C. For Problem C, use the prompt: 'We are working backwards — we know the area ratio is 25. What operation reverses squaring?' [WAIT 12 seconds.] Cold-call a learner who has written $\sqrt{25} = 5$ in their notebook. After the formula box is copied, say: 'Turn to your partner. Use this model to explain why filling the adult uji mug needed four small cups of water, not two. You have 2 minutes.' Cold-call 2 pairs. Affirm and connect: 'You have just explained the phenomenon that puzzled us in Lesson 1. $k = 2$, $k^2 = 4$ — four small cups. The mathematics predicted it perfectly.'	Worked-Example with Finger-Vote Checkpoints — the finger-vote forces every learner to commit to a choice (k, k^2, or k^3) before the answer is revealed, making the decision-point cognitively explicit. Connecting back to the uji-cup phenomenon at the end of the phase reinforces that the abstract formula is the explanation for the concrete mystery learners observed in Lesson 1.	Observable indicators: (1) During each finger-vote, teacher notes the ratio of correct to incorrect responses and adjusts pacing accordingly — if fewer than 70% of learners show the correct finger for Problem A, re-teach the area scale factor before proceeding to Problem B. (2) All learners have the formula box ($k \rightarrow$ lengths, $k^2 \rightarrow$ areas, $k^3 \rightarrow$ volumes) correctly copied in their notebooks before the DQB phase begins. Teacher spot-checks 6 notebooks (2 high, 2 mid, 2 low attainment) during the pair discussion.

	follows: 'Looking back at our uji cups from Lesson 1 — the toddler cup was scaled by $k = 2$ to make the adult mug, and by $k = 3$ to make the catering flask. Use our unified model to explain every surprising number we measured in Lesson 1.'			
Driving Question Board (DQB) Creation  VIDEO: Scale factors and area <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Area and Volume of Similar Solids Practice Source: CK-12  Search ARES for similar readings	Learners individually write one 'I now understand...' statement and one 'I still wonder...' question on separate sticky notes (or on torn notebook paper if sticky notes are unavailable). Two volunteers come to the front and update the class DQB: they add a new card titled 'The Unified Model' containing the formula summary ($k \rightarrow$ lengths, $k^2 \rightarrow$ areas, $k^3 \rightarrow$ volumes) and attach it to the central driving question card. They also move any previously posted 'mystery' questions (e.g., 'Why did we need 9 small cups to fill the flask?' from Lesson 1) into a new column labelled 'Explained by the Unified Model'. The class briefly reviews the DQB together — identifying which sub-questions of the driving question are now answered and which remain open (e.g., applying the model to non-cylindrical shapes, and to the cost of manufacturing similar containers).	Distribute sticky notes (or instruct learners to tear a page corner) and say: 'Write one thing you now understand about scaling — be specific, use numbers. And write one thing you are still wondering. You have 2 minutes.' [WAIT 2 minutes.] Invite volunteers for the DQB update — give them 2 minutes to arrange the cards while the class reads the board silently. Then facilitate a 3-minute whole-class scan of the DQB: 'Look at the board. Count how many of our original mystery questions from Lesson 1 can now be answered using the unified model.' Cold-call 3 learners to read one explained question and state the explanation in one sentence. Say: 'Our driving question asks why doubling a container makes it hold eight times as much. Can anyone answer that now in one sentence?' [WAIT 12 seconds.] Cold-call 2 learners. Acknowledge that the remaining open questions — particularly about designing and building containers — will be the focus of Lesson 7.	DQB Reorganisation with 'Explained' Column — physically moving mystery cards from 'open' to 'explained' gives learners a visible, tangible record of their growing understanding and makes the cumulative progress of the evidence-gathering sequence concrete. The remaining open questions sustain inquiry momentum into Lesson 7.	Observable indicator: Every learner has produced a specific, numerical 'I now understand...' statement (e.g., 'I now understand that $k = 3$ means the large tank needs 9 times more paint and holds 27 times more water') rather than a vague one. Teacher reads 8–10 sticky notes during the next phase to identify persistent misconceptions for the Lesson 7 opening hook.
Model Building Phase  VIDEO: Dilating triangles: find the error <i>Related topic: not an exact match</i>	Learners individually construct a Complete Annotated Model in their notebooks. The model has two components: (A) a labelled diagram showing the three uji cups (toddler, adult, catering flask) with annotated scale factors — all linear dimensions labelled with $\times k$	Say: 'You now have 12 minutes to build your complete annotated model — diagram plus formula box. This is your personal mathematical tool for everything that follows in this unit and in life. Make it clear enough that you could explain it to a younger	Annotated Diagram + Formula Box — combining a visual representation (the labelled uji-cup diagram) with a symbolic one (the formula box) exploits dual-coding to strengthen long-term retention. Using the anchoring phenomenon as the subject of the diagram	Observable indicators: (1) Exit Ticket answers: (a) $k = 4$, area scale factor = 16, so 16 times more metal; (b) volume scale factor = 64, so $64 \times 5 = 320$ litres; (c) cost proportional to area $\rightarrow 16 \times \text{Ksh } 800 = \text{Ksh } 12,800$. Teacher sorts exit tickets into three piles (all

for this lesson. Source: Khan Academy (English - US curriculum) Search ARES for similar videos  EXERCISE: Dilations: scale factor <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar readings	and $\times 3$, all surface-area labels showing $\times k^2$ and $\times 9$, and all volume/capacity labels showing $\times k^3$ and $\times 27$; (B) a Formula Summary Box containing: 'For two similar figures/solids with linear scale factor k: Length ratio = k, Area ratio = k^2, Volume ratio = k^3' — with the Rift Valley tank numbers ($k = 3$: area $\times 9$, volume $\times 27$) given as a worked example beneath the box. Learners who finish early add a second example: the uji cups with $k = 2$ (adult vs toddler) and $k = 3$ (flask vs toddler). The lesson closes with a 5-minute individual Exit Ticket: 'Two similar matumbo (cooking pots) used at a Nairobi nyama choma restaurant are similar in shape. The large pot has a height 4 times that of the small pot. (a) How many times more metal was needed to make the large pot? (b) If the small pot holds 5 litres of soup, how much does the large pot hold? (c) If the small pot cost Ksh 800 to manufacture, estimate the manufacturing cost of the large pot, assuming cost is proportional to the metal used.' Learners submit or show the Exit Ticket to the teacher at the door.	sibling.' [Play soft background music or set a silent working timer for 12 minutes.] Circulate and give specific praise: 'I can see you have labelled the volume arrow $\times k^3$ — excellent. Now add the actual numbers for $k = 3$ below it.' For learners who have drawn the diagram but not the formula box, prompt: 'Your diagram is strong. Now add the formula box — what are the three rules?' [WAIT 10 seconds before assisting.] At the 12-minute mark, say: 'Close your models. Here is your exit ticket — 5 minutes, individual work, no discussion.' [Distribute or display the nyama choma pot problem.] Collect tickets at the door. As learners leave, tell each one: 'In Lesson 7 we will use this model to design and build — you are ready.'	ensures that the model remains connected to the original observable mystery rather than becoming an abstract set of rules.	correct, partially correct, incorrect) before Lesson 7 and uses the results to plan targeted warm-up support. (2) All learner notebooks contain a complete annotated model with both diagram and formula box — checked during model-building circulation.
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D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) During the finger-vote checkpoints in the Explain phase, what percentage of learners correctly identified k^2 for area and k^3 for volume on the first attempt — and which of the three problems (A, B, or C) produced the most incorrect responses? What does this tell you about which relationship (area vs volume vs inverse) needs reinforcement in Lesson 7? (2) Review the exit-ticket piles: did learners who correctly solved Part (a) (area scale factor) also correctly solve Part (b) (volume scale factor), or did a subset succeed on area but fail on volume? This pattern would indicate that k^3 remains the weaker of the three relationships and should anchor the Lesson 7 opening. (3) Read the 'I still wonder...' sticky notes from the DQB phase: do any questions signal confusion about when to use k versus k^2 versus k^3 (i.e., confusion about which physical quantity determines which scale factor)? If so, design a 5-minute sorting warm-up for Lesson 7 where learners categorise quantities (length of pipe, litres of paint, area of label, number of bricks) under k , k^2 , or k^3 before the new content begins. (4) Were all learners able to connect the exit-ticket pot problem back to the uji-cup phenomenon, or did some treat it as an isolated calculation? If the latter, open Lesson 7 by explicitly asking learners to narrate the connection aloud.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Two similar cylindrical water tanks from the Rift Valley with linear scale factor $k = 3$; a unified $k / k^2 / k^3$ table completed and verified using digital devices; the GeoGebra/Kenya Education Cloud simulation showing surface area and volume changing as cylinder dimensions are scaled by $k = 2, 3, 4$, and 5 .
What did I learn?	For any two similar figures or solids with linear scale factor k : corresponding lengths scale by k , corresponding surface areas scale by k^2 , and corresponding volumes scale by k^3 . For $k = 3$: the large Rift Valley tank needs 9 times more paint ($k^2 = 9$) and holds 27 times more water ($k^3 = 27$). This single rule — one value of k encoding three scaling behaviours — explains every surprising number observed in the Shrinking Uji Cup phenomenon since Lesson 1.
How does this explain the phenomenon?	The anchoring phenomenon is now fully explained: when the uji cup manufacturer doubled every dimension ($k = 2$), the surface area of the label increased by $k^2 = 4$ (four times more paper) and the volume increased by $k^3 = 8$ (eight small cups needed to fill it, not two). When dimensions were tripled ($k = 3$), the label needed $k^2 = 9$ times more paper and the volume needed $k^3 = 27$ times more water (nine small cups, not three). The driving question — 'Why does doubling the size of a container make it hold eight times as much?' — is answered: because volume scales as the cube of the linear scale factor, so $k = 2$ gives $k^3 = 8$.

LESSON 7 (80 minutes (2 × 40-minute lessons)): Real-Life Applications and Problem Solving — Apply & Communicate

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners apply their accumulated understanding of linear, area, and volume scale factors to multi-step, real-life problem contexts — scale drawings, recipe scaling, and map reading — and publicly communicate a full, evidence-backed answer to the driving question.
Knowledge	Learners know that: (i) linear scale factor k scales lengths, (ii) area scale factor k^2 scales surface areas, and (iii) volume scale factor k^3 scales volumes; and that these relationships hold for ANY pair of similar figures or solids.
Skills	Learners can: (a) read and construct a scale drawing of a school building, (b) use volume scale factor to resize a recipe (e.g., a mandazi recipe scaled for a larger tin), (c) calculate real distances from a map using a stated scale, and (d) present one self-researched real-life example of similarity and enlargement to the class with supporting evidence.
Attitudes	Learners appreciate the coherence and utility of similarity and enlargement across diverse real-life domains (architecture, catering, cartography) and demonstrate confidence in communicating mathematical reasoning to peers.
Key Inquiry Question	How is similarity and enlargement applied in real life?
Purpose in Storyline	In Lessons 1–6, learners built the full mathematical model: they measured the uji cups, discovered the $k / k^2 / k^3$ surprise, derived the scale-factor rules, and applied them to solids. Lesson 7 is the 'Apply & Communicate' capstone: learners carry those rules out of the classroom and into scale drawings of school buildings, a Kenyan catering recipe, and a road map — then return to the Driving Question Board to articulate a complete, evidence-supported answer to the driving question.
Safety Notes	No physical hazards. If learners use mobile phones or LDDs to research additional examples, remind them to observe online safety guidelines (KICD PCI: Online Safety) — use school-approved sites only, do not share personal information, and report inappropriate content to the teacher.

B. LESSON OVERVIEW

Learners work in groups through three real-life application stations — a scale drawing task, a recipe-scaling task (mandazi dough for a school party), and a map-reading task using a Kenyan road map — each of which requires them to choose and apply the correct scale factor (k , k^2 , or k^3). Each group then researches one additional real-life example using print or digital resources and delivers a 2-minute presentation to the class. The lesson closes with a full-class return to the Driving Question Board, where learners compose and publicly defend a complete answer to the driving question, explicitly linking evidence from all seven lessons back to the anchoring phenomenon of the uji cups.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Scale factors and area	Learners sit in their established groups of 4–5. Each learner silently reads the 'Karibu Catering Challenge' prompt card placed on	Distribute prompt cards. Say: 'Read silently. Write your own prediction — no group discussion yet. You have 3 minutes.' [Set	Activation of Prior Knowledge (Prediction T-Chart): By committing to a written prediction before calculating, learners	Observable indicator: Each learner has a written prediction with a stated reason in their book before group discussion begins. The

Source: Khan Academy (English - US curriculum) Search ARES for similar videos HTML: Use Scale Factor When Problem Solving (Flexbook) Source: CK-12 Search ARES for similar readings	every desk: 'Mama Wanjiru bakes mandazi for her family using a small round tin of diameter 10 cm. For the Grade 10 end-of-term party she wants to use a similar tin of diameter 20 cm. She plans to mix DOUBLE the amount of dough. Will that be enough? Write your prediction and your reason before we do any calculation.' Learners write individual predictions in their exercise books (3 minutes). The teacher then cold-calls 4 learners — one per group — asking each to state their prediction and the rule they think applies. Predictions are recorded on the board in a T-chart: 'Prediction' 'Mathematical reason given.'	visible timer.] After 3 minutes: 'Pens down. Let us hear four predictions.' Cold-call Learner 1: 'Tell us your prediction and the one rule from our lessons that supports it.' [WAIT 12 seconds.] Cold-call Learner 2, 3, 4 in the same way. Record all predictions verbatim on the board without evaluating them. Say: 'We are going to test every one of these predictions today. Keep them in mind as we work.'	surface their current mental model of the k^3 rule. Misconceptions — such as 'double the diameter means double the dough' — become visible and productive; the teacher can target them explicitly during the Explain phase.	teacher scans 4–6 books during cold-calling. Red flag: any learner who predicts 'double the dough is enough' without connecting to a scale-factor rule — note these learners for targeted questioning during Station 2.
Observe Phase VIDEO: Scale drawing: centimeters to kilometers Source: Khan Academy (English - US curriculum) Search ARES for similar videos HTML: Use Unit Scale When Problem Solving (Flexbook) Source: CK-12 Search ARES for similar readings	Groups rotate through three evidence-gathering stations (approximately 8 minutes each, with a 30-second transition signal). Each station has a printed task card and any required materials. **Station 1 — Scale Drawing (Architecture):** Task card: 'The diagram shows a floor plan of the Mwangaza Primary School staff room. The scale is 1 : 50. (a) Measure the length and width of the room on the diagram in cm. (b) Calculate the actual length and width in metres. (c) A similar staff room at Jua Kali Secondary School is built with all dimensions scaled up by a factor of 1.5. What is the actual floor area of the Jua Kali staff room? How many times larger is it than the Mwangaza room?' Learners record all measurements and calculations on their station worksheet.	Before stations begin, say: 'Each station gives you a real-life problem. Your job is to decide: does this problem need k, k^2, or k^3 — and to show your reasoning, not just the answer. You have 8 minutes per station. I will circulate.' [Start timer.] Circulate continuously. At Station 2, target learners identified in Phase 1 who predicted 'double the dough': ask 'You said double — what is the linear scale factor here? What does that tell you about volume?' [WAIT 12 seconds. Do not answer; redirect to a peer if needed.] At Station 3, prompt: 'What changes when the scale of a map doubles — the distances, the areas, or both? How?' [WAIT 10 seconds.] After each 8-minute interval, give a 30-second warning signal ('Finish your current calculation and prepare to move'). After all rotations, say: 'Before we debrief, compare your Station 2 answer to your prediction from Phase 1. Did	Structured Evidence Stations with Decision-Point Scaffolding: Each station is designed so that learners must explicitly choose between k, k^2, and k^3 rather than following a formula blindly. The three different real-life contexts (architecture, catering, cartography) mirror the unit's central claim — that the same mathematical relationship governs all similar figures — thereby reinforcing the anchoring phenomenon through analogical reasoning across domains.	Observable indicators: (1) Station 2 worksheet shows the correct identification of $k = 2$ and volume scale factor = 8, with 4 000 g (8×500 g) of flour; (2) Station 3 worksheet shows correct conversion of map distance to km and correct calculation of new map area as $4 \times 12 = 48 \text{ cm}^2$; (3) Station 1 worksheet shows correct area scale factor of $1.5^2 = 2.25$. Teacher checks at least one group's worksheet per station during circulation. Red flag: any group using k instead of k^3 for the flour quantity in Station 2 — intervene immediately with a pointed question: 'Tell me exactly what the linear scale factor tells us about LENGTH. Now, what does it tell us about VOLUME?'

	Station 2 — Recipe Scaling (Mandazi for the Party): Task card: 'Mama Akinyi's small mandazi tin is a cylinder of diameter 10 cm and height 8 cm. The large party tin is similar, with diameter 20 cm. (a) State the linear scale factor k. (b) Mama Akinyi's recipe uses 500 g of flour to fill the small tin. How much flour is needed to fill the large tin? Show which scale factor (k, k^2, or k^3) you used and WHY. (c) If one packet of Blue Band margarine covers the curved surface of the small tin, how many packets cover the curved surface of the large tin?' Learners write working and answers on the worksheet. **Station 3 — Map Reading (Road Distance):** Task card: 'A printed road-map extract shows the route from Kisumu to Kericho. The map scale is 1 : 500 000. (a) Using a ruler, measure the map distance between the two towns. (b) Calculate the actual road distance in km. (c) A cartographer wants to redraw the map at a scale of 1 : 250 000. By what linear scale factor is the new map enlarged relative to the original? If the area of Lake Victoria on the original map is 12 cm², what will its area be on the new map?' Learners record measurements and calculations on the worksheet.	the mathematics confirm your prediction?		
Explain Phase  VIDEO: Solving a scale drawing word problem Source: Khan Academy (English - US curriculum)  Search ARES	**Part A — Station Debrief (12 minutes):** Each group nominates a spokesperson. The teacher facilitates a whole-class debrief of each station in turn. For Station 2, the teacher explicitly revisits the predictions from Phase 1. Learners who predicted 'double	Station debrief — call on the Station 2 spokesperson first: 'Tell us your answer and which scale factor you used for the flour.' [WAIT 10 seconds.] After the answer, redirect to a learner who predicted 'double': 'Amina, you predicted double. Looking at the	Cognitive Conflict Resolution + Analogical Bridging: The explicit return to the Phase 1 prediction that was wrong ('double the dough') creates productive cognitive conflict. The teacher then bridges this conflict directly to the anchoring phenomenon ('this is	Observable indicators: (1) During debrief, at least 3 of 4 groups correctly identify and justify the scale factor for each station; (2) During Part B, each group's Research Menu card has a named example, a stated scale factor, and an identification of k / k^2 / k^3 written

for similar videos EXERCISE: Construct scale drawings Source: Khan Academy (English - US curriculum) Search ARES for similar readings	the dough is enough' are asked to explain what they now know. The class agrees on and records the three correct answers in their books. **Part B — Research & Mini-Presentation Preparation (8 minutes):** Each group uses available print resources (textbooks, newspapers, printed reference sheets) or, where LDDs are available, a vetted digital resource (Kenya Education Cloud or teacher-approved site) to find ONE additional real-life example of similarity and enlargement that they have not yet studied in this unit. Groups prepare a 2-minute spoken presentation: they must name the context, state the linear scale factor, and explain whether the key quantity they are reporting scales by k, k^2, or k^3 — and why. Suggested domains (printed on a 'Research Menu' card): architecture, fashion/tailoring, photography/printing, mapmaking, cooking/baking, packaging design, medical imaging.	mathematics now — what happened?' [WAIT 12 seconds.] Validate the moment of surprise: 'This is EXACTLY what happened with the uji cups on Day 1. The cup was twice as tall — but held eight times as much. That was not a mistake. That is k^3.' Write on the board: '$k = 2 \rightarrow$ Volume scale factor = $2^3 = 8 \rightarrow$ Need $8 \times 500 \text{ g} = 4\,000 \text{ g}$ of flour.' For Station 3, cold-call a different group: 'What linear scale factor relates the two maps?' [WAIT 10 seconds.] After the correct answer of $k = 2$, ask: 'So the area of Lake Victoria on the new map will be...?' [WAIT 10 seconds. Cold-call.] Confirm: '$k^2 = 4$, so $4 \times 12 = 48 \text{ cm}^2$.' For Part B, say: 'You have 8 minutes. Use your Research Menu card. Every person in the group must be able to explain the example — not just the presenter. Choose your presenter now.' Circulate, checking that groups are identifying which scale factor applies to their chosen example. Prompt hesitant groups: 'Is your example about length, surface area, or volume? That tells you which rule to apply.'	exactly what happened with the uji cups'), helping learners consolidate the generalised rule rather than treating each application as a separate case. The Research Menu card extends analogical bridging further across new domains.	before presentations begin. Red flag: a group that cannot articulate why their chosen example uses k^2 rather than k or k^3 — intervene with: 'Describe the quantity you are scaling. Is it a length, an area, or a volume? Now match it to the correct rule.'
Driving Question Board (DQB) Creation VIDEO: Scale factors and area Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE:	**Part A — Mini-Presentations (6 minutes):** Each group delivers their 2-minute presentation to the class. The class listens and on a sticky note writes one sentence: 'This example shows [scale factor type] because...' The sticky notes are added to the DQB under a new column headed 'Our Real-Life Evidence — Lesson 7.' **Part B — DQB Full Revisit (4 minutes):** The teacher draws the	For presentations, use a visible 2-minute timer per group. After each presentation, ask the class: 'Which scale factor did they use — k, k^2, or k^3? Write it on your sticky note and add it to the DQB.' For the DQB full revisit, stand at the board and sweep your hand across all columns: 'Look at how far we have come. Every column answers part of the driving question. Now I want you to write ONE complete answer — use evidence from at least three	Public Knowledge Construction via DQB Synthesis: The Driving Question Board functions as a visible, cumulative record of the class's collective epistemic journey. By physically viewing all seven lessons' evidence simultaneously and composing a written synthesis, learners perform the metacognitive act of consolidation — moving from fragmented observations (the uji cup surprise) to a generalised,	Observable indicators: (1) Each group's sticky note correctly names a real-life domain, a scale factor type, and a brief reason; (2) In cold-called written answers, learners explicitly name k, k^2, k^3, reference at least one concrete example (e.g., the uji cups or the mandazi tin), and state the general rule. Red flag: a written answer that only restates the rule without evidence — prompt: 'Which of our seven lessons gave you the data

Reading Maps, Legends/Scale Practice Source: CK-12  Search ARES for similar readings	class's attention to the entire DQB accumulated across all 7 lessons. Learners look at every column — from 'What we noticed about the uji cups' (Lesson 1) through to 'Our Real-Life Evidence' (today). The teacher poses the driving question in full: 'Why does doubling the size of a container make it hold eight times as much — and how can we use this mathematical pattern to design, build, and predict the properties of any similar shape or object?' Learners silently draft a complete written answer in their books (3 minutes). One learner is cold-called to read their answer aloud.	of our lessons, including today's.' [WAIT 3 minutes writing time. Do not narrate; allow silence.] Cold-call: 'Ochieng, read your answer.' [WAIT 10 seconds before prompting.] After the reading, ask the class: 'What evidence from today's stations would you add to Ochieng's answer?' [Cold-call 2 more learners.]	transferable mathematical principle. The mini-presentations add peer-generated evidence to the board, broadening the evidential base beyond teacher-curated examples.	that proves this rule? Mention it.'
Model Building Phase  VIDEO: Scale drawing: centimeters to kilometers Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Use Scale Factor When Problem Solving (Flexbook) Source: CK-12  Search ARES for similar readings	Each learner opens their 'Similarity & Enlargement Model' — the annotated diagram they have been building since Lesson 1 (originally a sketch of the three uji cups with arrows and labels). Today they complete a final revision: (a) They add a new outer ring or banner to their model titled 'Where This Works in Real Life' and populate it with at least FOUR real-world domains identified across the lesson sequence (e.g., scale drawings, recipe sizing, map reading, packaging, photography). (b) They write or confirm the three general rules in symbolic and word form inside the model: 'If linear scale factor = k, then area scale factor = k^2 and volume scale factor = k^3.' (c) They annotate the uji cup diagram — still at the centre of the model — with today's mandazi example: 'e.g., mandazi tin: $k = 2 \rightarrow$ need $k^3 = 8$ times the dough.' (d) Learners exchange models with a partner, who checks the	Say: 'Your model has grown with us since Lesson 1 — from a puzzling cup of uji to a full mathematical theory. Today you give it its final layer. Open your models. You have 10 minutes to complete parts (a), (b), and (c).' [Set timer.] Circulate, checking that the three rules are symbolically correct (k, k^2, k^3) and that the real-life domains listed are genuinely diverse. If a learner has only listed cooking examples, prompt: 'What did Station 1 add to your model? What about Station 3?' [WAIT 10 seconds.] After 10 minutes: 'Exchange with a partner. Peer-review for 2 minutes — check the three rules and count the domains. Sign if satisfied.' After peer-review: cold-call one learner to read their 'Key Insight' sentence aloud. Close with: 'From one puzzling cup of uji to a rule that works for every similar shape on earth — that is the power of mathematical modelling. Well done.'	Cumulative Annotated Model Revision (Final Synthesis): The learner's personal, evolving model — anchored throughout by the uji cup diagram — serves as a cognitive artefact that makes the growth of understanding visible and concrete. Adding the 'real-life domains' outer ring in Lesson 7 completes the model's transformation from a description of one phenomenon to a generalisable scientific and mathematical principle. Peer-review introduces a low-stakes accountability structure that reinforces precision in symbolic notation.	Observable indicators: (1) Completed model contains the three rules in both symbolic and word form (k, k^2, k^3); (2) At least four distinct real-life domains are listed in the outer ring; (3) The uji cup / mandazi annotation is present and numerically correct; (4) Partner's signature is present. Red flag: model still has only the uji cup diagram without any real-life extensions — this learner needs a one-on-one follow-up at the start of Lesson 8 to consolidate generalisation before the summative assessment.

	three rules are correctly stated and that at least four real-life domains are listed, then signs the model as 'peer-reviewed.' Learners who finish early write a 'Key Insight' sentence at the bottom of their model: 'The most surprising thing this model shows is...'			
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D. TEACHER REFLECTION

After the lesson, consider the following questions to guide your preparation for Lesson 8 (the unit summative assessment):

1. During the Phase 1 T-chart, what proportion of learners predicted 'double the dough is enough'? If more than one-third held this misconception entering Lesson 7, build a 5-minute warm-up into Lesson 8 that revisits the k^3 rule with a new, rapid example before the assessment begins.
2. In the station debrief, were learners able to articulate WHY they chose k vs. k^2 vs. k^3 — or did they apply the correct rule without being able to justify it? Justification is assessed at the 'Meets Expectations' and above level; if learners can calculate but not explain, provide a sentence-starter scaffold in the assessment: 'I chose this scale factor because...'
3. Which group's mini-presentation showed the most original or surprising real-life example? Consider displaying their example as an extension challenge on the Lesson 8 assessment paper.
4. Review the peer-reviewed models before Lesson 8. Any model missing the symbolic rules (k , k^2 , k^3) or with fewer than four real-life domains signals a learner who may need additional support during the assessment. Plan a differentiated prompt card for these learners.
5. Did the DQB synthesis activity produce written answers of sufficient depth? If learners' written answers referenced only one lesson's evidence, plan a brief 'evidence carousel' at the start of Lesson 8 where each group shares their strongest piece of evidence from the unit before the summative task begins.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners measured scale drawings of a school building, calculated flour quantities for a scaled-up mandazi tin ($k = 2$, requiring $8\times$ the dough), and used a Kisumu–Kericho road map at scale 1:500 000 to calculate real distances and compare map areas at a different scale.
What did I learn?	The linear scale factor k governs lengths, k^2 governs areas (including surface area and map area), and k^3 governs volumes (including recipe quantities for similar containers). These three rules apply consistently across all real-life domains — architecture, catering, cartography — involving similar figures or solids.
How does this explain the phenomenon?	The uji cup phenomenon — where doubling every dimension of the cup requires $8\times$ as much water to fill it, not $2\times$ — is explained by the k^3 volume rule. Because volume depends on three linear dimensions multiplied together, scaling each by k scales the total volume by $k \times k \times k = k^3$. The same logic explains why Mama Akinyi needed 4 000 g of flour (not 1 000 g) for the large mandazi tin, and why the Lake Victoria area on a redrawn map at twice the linear scale is $4\times$ larger, not $2\times$.

LESSON 8 (80 minutes): Project Presentations, Final Explanation & DQB Completion

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners synthesise all learning from the unit by presenting their similar-container projects, completing the Driving Question Board, comparing their Day-1 mental model with their final model, and demonstrating mastery of linear, area, and volume scale factors through a written class activity.
Knowledge	Properties of similar figures; relationships between linear scale factor k , area scale factor k^2 , and volume scale factor k^3 ; centre of enlargement; application of similarity and enlargement to real-life design.
Skills	Measuring and recording dimensions of self-constructed similar containers; calculating and verifying linear, area, and volume scale factors; presenting mathematical reasoning clearly to peers; evaluating peers' work using a rating scale; revising and completing a cumulative explanatory model.
Attitudes	Appreciation of the utility and aesthetic value of similarity and enlargement in everyday design; responsibility in taking care of constructed models and instruments; confidence in communicating mathematical ideas; respect for peers' contributions and different approaches.
Key Inquiry Question	How is similarity and enlargement applied in real life?
Purpose in Storyline	In Lesson 1 learners were puzzled that doubling the height of the uji cup did not double — but quadrupled — the water it held. Over seven lessons they built the mathematical tools to explain this. In Lesson 8 they close the loop: they present containers they made themselves, prove the $k / k^2 / k^3$ pattern with real measurements, answer every sub-question on the DQB, and write a final individual explanation of the anchoring phenomenon — demonstrating full competency.
Safety Notes	Learners should handle cutting tools (scissors, craft knives) safely when displaying containers. Cardboard edges can be sharp — handle carefully. If glue guns were used in construction, remind learners these are not to be brought into the presentation room.

B. LESSON OVERVIEW

This is the final lesson of the Similarity and Enlargement unit. It is structured in three connected acts. First (Predict → Observe, ~25 min), learner groups present their self-constructed similar containers — cubes, cuboids, or cylinders made from locally available materials in at least two sizes — and peers evaluate each presentation using a rating scale. Second (Explain → DQB Completion, ~30 min), the whole class conducts a guided final explanation connecting every piece of evidence gathered across Lessons 1–7 back to the anchoring phenomenon (the uji cups), and every sub-question on the Driving Question Board is formally answered and marked. Third (Model Building → Written Assessment, ~25 min), learners individually revise their cumulative model diagram and complete a written class activity that assesses all specific learning outcomes. The lesson culminates with learners articulating — in writing — a complete, mathematically precise answer to the driving question: "Why does doubling the size of a container make it hold eight times as much?"

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase	Learners retrieve their Lesson-1 written prediction cards (stored in	Before presentations begin, display the three uji cups (or	Retrieval practice with phenomenon anchoring — by	Observable indicator: each learner's written sentence correctly

 VIDEO: Scale factors and area Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Dilations: scale factor <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	their portfolios) about why the large uji cup needed nine small cups of water to fill it, not three. Working individually for 3 minutes, each learner writes one sentence completing the stem: 'I now know the large uji cup holds nine times as much water because...' They then turn and share with a partner. Selected pairs share with the whole class. Learners also retrieve their Day-1 rough model sketch and place it on their desk alongside the final model page they have been building since Lesson 1.	photographs) at the front — exactly as in Lesson 1. Say: 'We started with a mystery. Before your classmates show us what they built, let's check what you now know.' Distribute the Lesson-1 prediction cards back to learners. Give the 3-minute individual writing time. Then say: 'Turn and tell your partner your sentence.' Allow 90 seconds pair talk. Cold-call 4 learners (aim for gender balance and mixed ability): 'Aisha, please read your sentence.' After each response, WAIT 12 seconds before commenting. Prompt with: 'Can you name the exact scale factor that explains that?' If a learner says 'because the cup is bigger,' push with: 'How much bigger — in area? In volume? What is the mathematical relationship?' Record 3–4 key phrases on the board under the heading 'What we now know'. These will anchor the final explanation later.	reconnecting to the Lesson-1 uji-cup mystery before presentations, learners activate the full conceptual arc of the unit in working memory, priming them to evaluate peers' projects against the core mathematical ideas (k, k^2, k^3) rather than purely aesthetic criteria.	names at least one of the three scale factor relationships (linear k, area k^2, volume k^3) and connects it to the uji-cup context. Teacher scans sentences during pair-share; learners who write only qualitative statements ('because the cup is bigger') are noted for targeted questioning during the Explain phase.
Observe Phase  VIDEO: Scale factors and area Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Dilations: scale factor <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar	Each group (3–4 learners) presents their similar-container project for 4–5 minutes. The presenter must: (1) display the two (or more) containers side by side; (2) read aloud at least two corresponding measurements (e.g., height and diameter for cylinders, or length and width for cuboids); (3) state the linear scale factor k with working shown; (4) calculate the area scale factor k^2 and the volume scale factor k^3 and compare them to actual measured or calculated surface area and volume; (5) describe one challenge faced during construction and how it was resolved. While each group presents, every other learner	Seat the class in a horseshoe so all containers are visible. Before the first presentation say: 'Your job as an audience is to be a mathematical critic — not just to admire the container. Use your rating scale. Does their k^2 match the area ratio? Does k^3 match the volume ratio? If not, that is an important finding, not a mistake to hide.' Time each group strictly (use a visible timer). After each presentation, invite one audience question: 'Who has a wonder question for this group?' WAIT 10 seconds. Cold-call if no volunteer. After all groups have presented, ask the whole class: 'Which group's containers showed the k / k^2 / k^3 relationship most clearly —	Evidence-to-model presentation — learners use their own constructed artefacts as primary evidence, making the abstract relationship $k \rightarrow k^2 \rightarrow k^3$ concrete and verifiable. Peer rating shifts evaluation authority to learners and requires them to apply the mathematical criteria rather than passively receive information. The 'warm/wonder' protocol promotes collaborative, respectful mathematical discourse.	Observable indicators: (1) Completed peer rating scales — teacher collects these after presentations to identify which criteria were consistently rated low across groups. (2) Quality of wonder questions — questions that reference specific scale factors (e.g., 'Your k was 2 but your volume ratio was 6 — can you explain?') indicate deeper understanding than aesthetic questions. (3) Teacher notes any group whose stated k^2 or k^3 values are inconsistent with their measurements — these become the focus of targeted correction in Phase 3.

readings	completes a Peer Rating Scale (four criteria, each scored 1–4): (a) Similar shape clearly demonstrated; (b) Linear scale factor correctly calculated and stated; (c) Area and volume scale factors correctly calculated; (d) Challenge documented and explained. Audience members write one 'warm' comment and one 'wonder' question on a sticky note to hand to the presenting group afterward.	and how do you know?' WAIT 15 seconds. Cold-call 3 different learners. If a group's calculations contain errors, do not correct immediately — note it on the board as 'Let's investigate this together in Phase 3.' For groups presenting cylinders, prompt: 'Your cylinder is just like our uji cups — what does your data say about why the big cylinder holds k^3 times as much?'		
Explain Phase  VIDEO: Dilations: center <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Dilations: scale factor <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	The teacher leads a whole-class final explanation structured around the uji cups. Learners open their exercise books to a fresh page headed 'Final Explanation'. The teacher works through a summary example on the board using measurements from one of the presented containers (or the uji cups), and learners write their own version simultaneously. The explanation covers: (a) definition and properties of similar figures; (b) determining the centre of enlargement; (c) linear scale factor k from corresponding lengths; (d) area scale factor = k^2 and why (two dimensions each scaled by k, so area scales by $k \times k$); (e) volume scale factor = k^3 and why (three dimensions each scaled by k, so volume scales by $k \times k \times k$); (f) a worked example: if a small cylindrical uji cup has radius 4 cm and height 8 cm, and the large cup has radius 8 cm and height 16 cm, find k, the surface area of each cup, and the volume of each cup — then verify that the area ratio equals k^2 and the volume ratio equals k^3. Any calculation errors noted during presentations are resolved collaboratively here: the teacher writes the erroneous result	Begin: 'Let's use what every group showed us to write the most complete explanation we have ever written for this unit.' Work through the uji-cup example step by step on the board, pausing after each step: 'Write this in your own words before I continue.' For the mandazi-mould extension: 'In your book, work out k, k^2, and k^3 for the mandazi moulds. You have 4 minutes.' WAIT for independent work. Cold-call 3 learners to share answers, choosing one who has the correct answer, one who has a partial answer, and one who made an error: 'Kamau, show us your k. Wanjiru, yours was different — walk us through your thinking.' Address any presentation errors from Phase 2 here: 'Group 3 got a volume ratio of 6 when k was 2. Whose volume ratio was different? Let's resolve this together.' Use a targeted question: 'If $k = 2$, how many times does k appear when we scale volume — think about length, width, height.' WAIT 12 seconds. Do not answer for learners — redirect to a peer if needed.	Collaborative error analysis with guided note-taking — by deliberately surfacing calculation errors from presentations and resolving them collaboratively, the teacher models mathematical self-correction and deepens procedural understanding. The mandazi-mould extension transfers the concept to a new Kenyan context, testing flexible application beyond the original phenomenon.	Observable indicators: (1) Learners' 'Final Explanation' notes — teacher circulates and stamps or ticks books where the uji-cup worked example is correctly completed with all three scale factors verified. (2) Mandazi-mould extension — correct answers: $k = 3$, $k^2 = 9$, $k^3 = 27$; learners who obtain $k^3 = 27$ but cannot explain why demonstrate procedural fluency without conceptual depth — note for written assessment. (3) Learner responses to error-resolution questions — learners who can identify that 'volume uses three dimensions so k appears three times' are demonstrating the target conceptual understanding.

	on the board and asks 'What should this be? How do we correct it?' Learners also apply the concept to a new Kenyan context: a small mandazi mould has diameter 5 cm; a large similar mould has diameter 15 cm. What are k, k^2, and k^3? How many small moulds of oil would fill the large mould?			
Driving Question Board (DQB) Creation  VIDEO: Dilating triangles: find the error <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Dilations: scale factor <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	The class Driving Question Board — maintained since Lesson 1 — is displayed at the front (or projected). It contains the main driving question and all sub-questions learners posted across the unit. The teacher reads each sub-question aloud. For each one, the class decides together: (a) Has this question been fully answered? (b) What is the answer, in one or two precise sentences? A nominated learner writes the agreed answer on a green sticky note and places it over the original question card, completing the DQB. Typical sub-questions that should now be answerable include: 'Why does the area not just double when the lengths double?'; 'What is a scale factor?'; 'Why do architects use scale models?'; 'Is a shadow of a shape similar to the original?'; 'How do manufacturers make containers of different sizes?'; 'What is the centre of enlargement?' If any sub-question remains unanswered, learners note it as an extension question for self-study. Finally, learners re-read the main driving question — 'Why does doubling the size of a container make it hold eight times as much — and how can we use this mathematical pattern to design, build, and predict the properties of any	Stand beside the DQB. 'This board has been our compass since Lesson 1. Every question on it was raised by you. Let's honour that by answering every single one.' Read each sub-question clearly. For each: 'Who can answer this in one precise mathematical sentence? Hands up.' WAIT 10 seconds. If no hand, say: 'Think back to the lesson where we first found evidence for this.' After a learner responds, ask: 'Does everyone agree? Thumbs up / thumbs sideways / thumbs down.' For disagreements, facilitate briefly: 'Two different answers — which one uses the correct mathematical language?' For the final driving question vote, cold-call one Yes voter and one learner who hesitated: 'Tell me the part you are still unsure about.' Use this to bridge to the written assessment: 'The written activity will give you a chance to show me in detail exactly where you are with all of this.'	Driving Question Board synthesis — a public, collaborative closure of the inquiry arc. By requiring precise one-sentence answers (not vague summaries), the protocol forces learners to consolidate knowledge into transferable propositions. The vote on the driving question surfaces residual uncertainty and frames the written assessment as meaningful rather than arbitrary.	Observable indicators: (1) Quality of green-sticky-note answers — well-formed answers name the scale factor algebraically (e.g., 'Area scales by k^2 because two linear dimensions are each multiplied by k'). Vague answers ('because it gets bigger') signal incomplete understanding and are sent back for revision before being posted. (2) Thumbs vote during discussion — widespread thumbs-sideways on a specific sub-question alerts the teacher to a concept that may need brief re-teaching before or during the written activity. (3) DQB is now fully green — all sub-questions answered; any remaining orange cards are recorded by the teacher for follow-up.

	similar shape or object?' — and vote by show of hands: 'Can we now answer this fully? Yes or No?' The teacher asks two learners who voted Yes to give their answer aloud.			
Model Building Phase  VIDEO: Dilating shapes: shrinking <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos  EXERCISE: Dilations: scale factor <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar readings	Learners first spend 5 minutes revising their cumulative model diagram (built progressively since Lesson 1). They add a final layer: a labelled arrow from 'linear scale factor k' branching into 'area scale factor k²' and 'volume scale factor k³', each with a real-number example from their own project. They also annotate one feature of their diagram that was wrong or incomplete in Lesson 1 and write beside it: 'I used to think... Now I know...'. They then complete a written class activity (individual, 15 minutes) comprising the following tasks: (1) Two similar cuboid water tanks used on Rift Valley farms — Tank A: 50 cm × 40 cm × 30 cm; Tank B has all dimensions scaled by k = 2.5. Find (a) the dimensions of Tank B, (b) the surface area of each tank and the area scale factor, (c) the volume of each tank and the volume scale factor, (d) the number of 20-litre jerricans needed to fill Tank B if Tank A holds exactly 2 jerricans. (2) A label designer in Nairobi wraps a cylindrical tin of radius 6 cm and height 10 cm. A similar larger tin has radius 9 cm. Find the linear scale factor, the area of the label for each tin (curved surface area only), and verify the area ratio equals k². (3) Short answer: In your own words, explain why doubling all dimensions of a container multiplies its volume by 8, not 2. Use the uji-cup story in your answer.	Distribute the written activity sheet. Say: 'This is individual work. Your model diagram is also individual. You may look at your notes and Final Explanation page.' Circulate throughout. After 8 minutes, give a quiet prompt: 'If you are on question 1(d), check: how many jerricans does Tank B hold? Remember k³.' Do not give answers — redirect with questions. With 3 minutes remaining, say: 'Finish question 3 even if question 2 is incomplete — your explanation in words is as important as your calculations.' Collect activity sheets. Before dismissal, ask the class to hold up their model diagrams: 'Look at your Lesson-1 sketch and your final model side by side. What is the single biggest change in your thinking?' Cold-call 3 learners for a one-sentence response. Close with: 'The uji cups were not a mystery — they were mathematics. You now own that mathematics.' Remind learners to store their constructed containers safely as they may be used as teaching resources.	Model revision with 'I used to think / Now I know' metacognitive annotation — this explicitly surfaces conceptual change, helping learners consolidate the difference between their naive understanding (scaling is linear) and their final scientific model (scaling is dimensional: k, k², k³). The written assessment is deliberately multi-representational: numerical calculation, formula application, and explanatory prose — mirroring the KICD assessment rubric levels.	Observable indicators: (1) Model diagrams — a complete final model shows $k \rightarrow k^2 \rightarrow k^3$ with a real-number example and a correctly annotated 'I used to think / Now I know' statement. (2) Written activity — mark scheme: Q1(a) 125 cm × 100 cm × 75 cm; Q1(b) SA of A = 9,400 cm², SA of B = 58,750 cm², ratio = 6.25 = k²; Q1(c) Vol A = 60,000 cm³, Vol B = 937,500 cm³, ratio = 15.625 = k³; Q1(d) Tank A holds 2 jerricans so Tank B holds 2 × 15.625 = 31.25 ≈ 32 jerricans (rounding up); Q2 k = 1.5, label A = 120π cm², label B = 270π cm², ratio = 270π/120π = 2.25 = k²; Q3 — assessed on: names k = 2, states each of 3 dimensions doubles so volume multiplies 2 × 2 × 2 = 8, connects explicitly to the uji cup. (3) Exceeds-expectation indicator: learner in Q1(d) explains that partial jerricans are not practical and rounds up, demonstrating real-life application judgment.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions before planning any remediation or extension:

1. **PROJECT QUALITY** — Did all groups correctly state and verify all three scale factors (k , k^2 , k^3) using their own measurements? If more than two groups had incorrect volume scale factor calculations, plan a 10-minute consolidation activity in the next available lesson using fresh examples from Kenyan packaging (e.g., Bidco cooking oil tins in 500 ml, 1 L, and 2 L sizes).
2. **DQB GAPS** — Were there any sub-questions on the DQB that the class could not answer confidently? Note these and include them as starter questions in the next strand's first lesson to reinforce cross-topic connections.
3. **WRITTEN ASSESSMENT** — Review the collected activity sheets before the next lesson. Tally the number of learners who: (a) correctly completed all three scale factor calculations; (b) correctly completed Q1 and Q2 but could not write the Q3 explanation in their own words; (c) completed fewer than two questions successfully. Group (b) may need guided writing support; Group (c) may require a structured re-teach of the k / k^2 / k^3 relationships before the sub-strand is formally closed.
4. **MODEL PROGRESSION** — Compare the Lesson-1 sketches and final models. Were the 'I used to think / Now I know' annotations specific and mathematical, or vague? Specific annotations (e.g., 'I used to think doubling size doubled volume; now I know volume scales by $k^3 = 8$ ') indicate deep conceptual change. Vague annotations signal that the learner has procedural fluency but has not fully restructured their prior understanding — follow up with a one-on-one conversation.
5. **EQUITY OBSERVATION** — During presentations, note which learners were consistently silent or rated by peers as below expectations. Were there learners whose constructed containers showed strong practical skill but weak written calculation? These learners may benefit from oral assessment as a supplementary approach.
6. **PHENOMENON CLOSURE** — Did every learner articulate — at least once, orally or in writing — a complete answer to the driving question that named k , k^2 , and k^3 ? The uji-cup mystery should feel fully resolved. If learners still express surprise or uncertainty about the nine-small-cups result, this is a signal that the volume scale factor conceptual link was not sufficiently consolidated across the unit.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners observed their own peer-constructed similar containers (cubes, cuboids, cylinders) made from locally available materials in at least two sizes. They measured corresponding dimensions side by side, saw with their own hands that every length scaled by the same factor k , and verified (or discovered errors in) their calculated surface area ratios and volume ratios. They also re-observed the three uji cups from Lesson 1 — now able to explain precisely why nine small cups of water, not three, are needed to fill the large cup.
What did I learn?	Learners consolidated that: (1) similar figures have equal corresponding angles and proportional corresponding sides; (2) the linear scale factor k is the ratio of any pair of corresponding lengths; (3) the area scale factor is always k^2 — because area involves two linear dimensions each scaled by k ; (4) the volume scale factor is always k^3 — because volume involves three linear dimensions each scaled by k ; (5) these relationships apply universally to all similar figures and solids, including the containers they designed and built themselves; (6) real-life applications include container manufacturing, architectural scale models, map-making, and packaging design.
How does this explain the phenomenon?	Learners explained — through project presentations, peer rating, collaborative DQB closure, final model annotation, and individual written tasks — why doubling all linear dimensions of any container multiplies its surface area by 4 ($= 2^2$) and its volume by 8 ($= 2^3$). They connected this explanation back to the anchoring phenomenon: the large uji cup has $k = 3$ relative to the small cup, so its cross-sectional area is $k^2 = 9$ times larger (explaining why nine small cups fill it) and its volume is $k^3 = 27$ times larger. They demonstrated this understanding using their own constructed containers and novel Kenyan contexts (Rift Valley water tanks, Nairobi label design, mandazi moulds), satisfying the KICD key inquiry question: How is similarity and enlargement applied in real life?

DIFFERENTIATION AND INCLUSION		
Learner Need	Support Strategies	Extension Strategies
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.